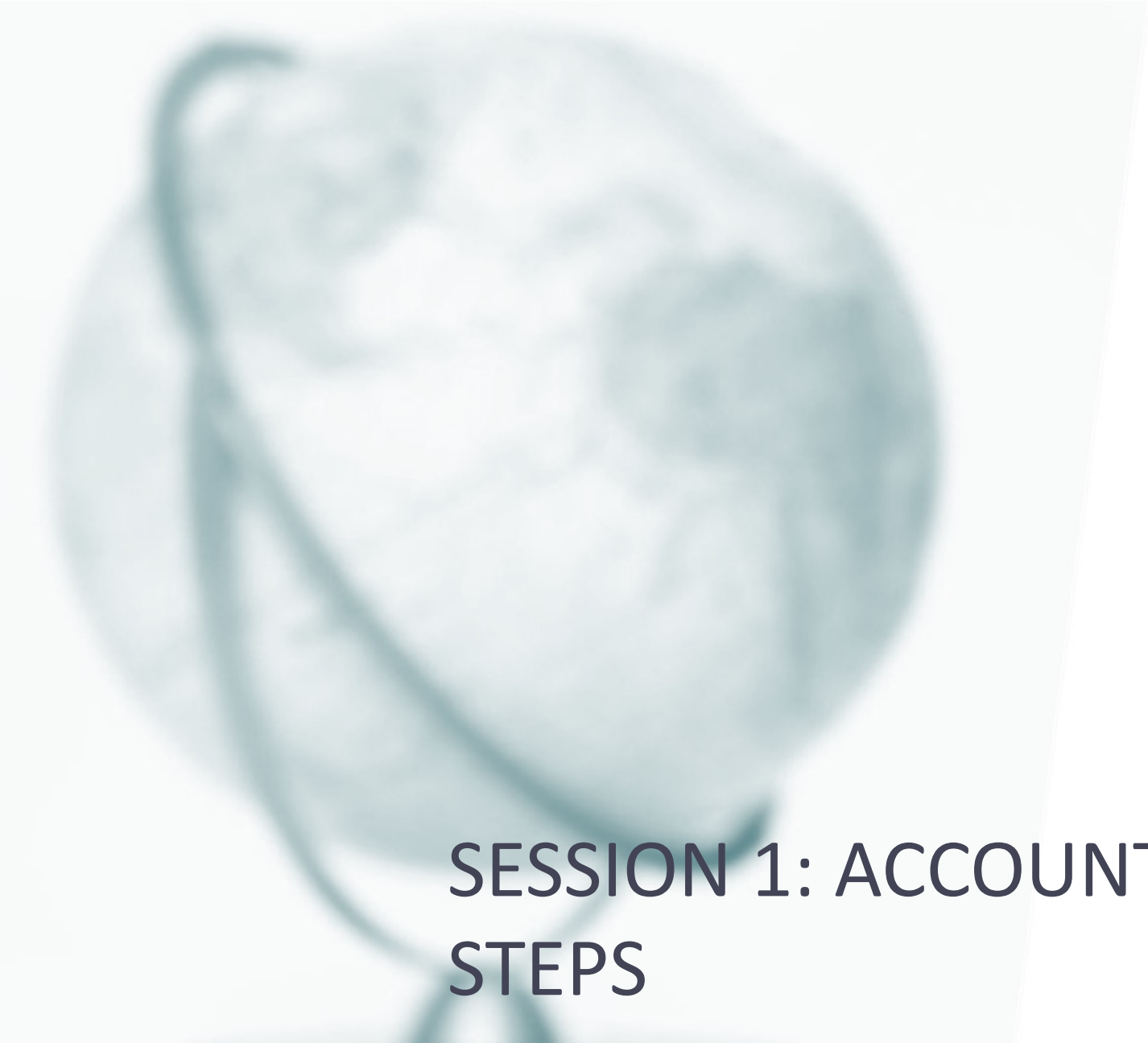




SESSION 1: ACCOUNTING FIRST STEPS

Accounting for Finance

The Accountant's Role

- In my view (and many accountants will disagree), it is the role of accounting
 - To check transactions and operations, as they occur
 - To record them in a consistent manner
 - To report the results in standardized form
- Much as accounting wants to makes itself more relevant and central to businesses, it is not the role of accounting:
 - Forecast the future, no matter how tempted.
 - Value assets or operations.
- **Bluntly put, an accountant is a historian, chronicling events that have already occurred, not predicting the future.**

The Accounting Questions..

- *What do you own?* List out the assets that a business has invested in, and how much it spent on those investments and *perhaps what these assets are worth today.*
- *What do you owe?* Specify the contractual commitments that a business has to meet, to stay in business. Simply put, this should include all borrowings, but is not restricted to those.
- *How much money did you make?* Measure the profitability of the business, both with accounting judgments on expenses, and based upon cash in and cash out.

The Accounting Statements

- **The balance sheet**, which summarizes what a firm owns and owes at a point in time, as well as an estimate of what equity is worth (through accounting eyes).
- **The income statement**, which reports on how much a business earned in the period of analysis, while providing detail on revenues and expenses.
- **The statement of cash flows**, which reports on cash inflows and outflows to the firm during the period of analysis and allows for a measure of cash earnings (as opposed to accounting earnings) and cash flows.

1. Balance Sheet

<i>Balance Sheet</i>			
Assets		Liabilities	
Long Lived Physical Assets	Fixed Assets	Current Liabilities	Short term obligations
Short Lived Assets	Current Assets	Debt	Long term debt
Investments in Securities & other business	Financial Assets	Other Liabilities	Other long term obligations
Assets which are not physical	Intangible Assets	Equity	Shareholders' Equity

2. Income Statement

	<i>Item</i>	<i>Explanation</i>
<i>Start with</i>	Revenues	Accountant's estimate of the revenues/sales generated by any transactions made the business during the period.
<i>Net out</i>	Cost of Goods Sold	Expenses associated with producing products or services that represent top line sales
To get	Gross Profit	Production profitability
<i>Net out</i>	Other Operating Expenses	Includes selling, general and administrative and other expenses associated with operations, but not directly tied to producing goods and services
To get	Operating Profit	Profitability of business
<i>Net out</i>	Financial Expenses	Expenses associated with the use of non-equity capital, especially debt.
To get	Taxable Income	Income for equity investors, prior to taxes
<i>Net out</i>	Taxes	Taxes due on taxable income
To get	Net Income	Income for equity investors, after taxes

3. Statement of Cash Flows

Statement of Cash Flows

Net cash flow from operations, after taxes and interest expenses

Cash Flows From Operations

Includes divestiture and acquisition of real assets (capital expenditures), disposal and purchase of financial assets and cash spent on acquisition of other firms.

+ Cash Flows From Investing

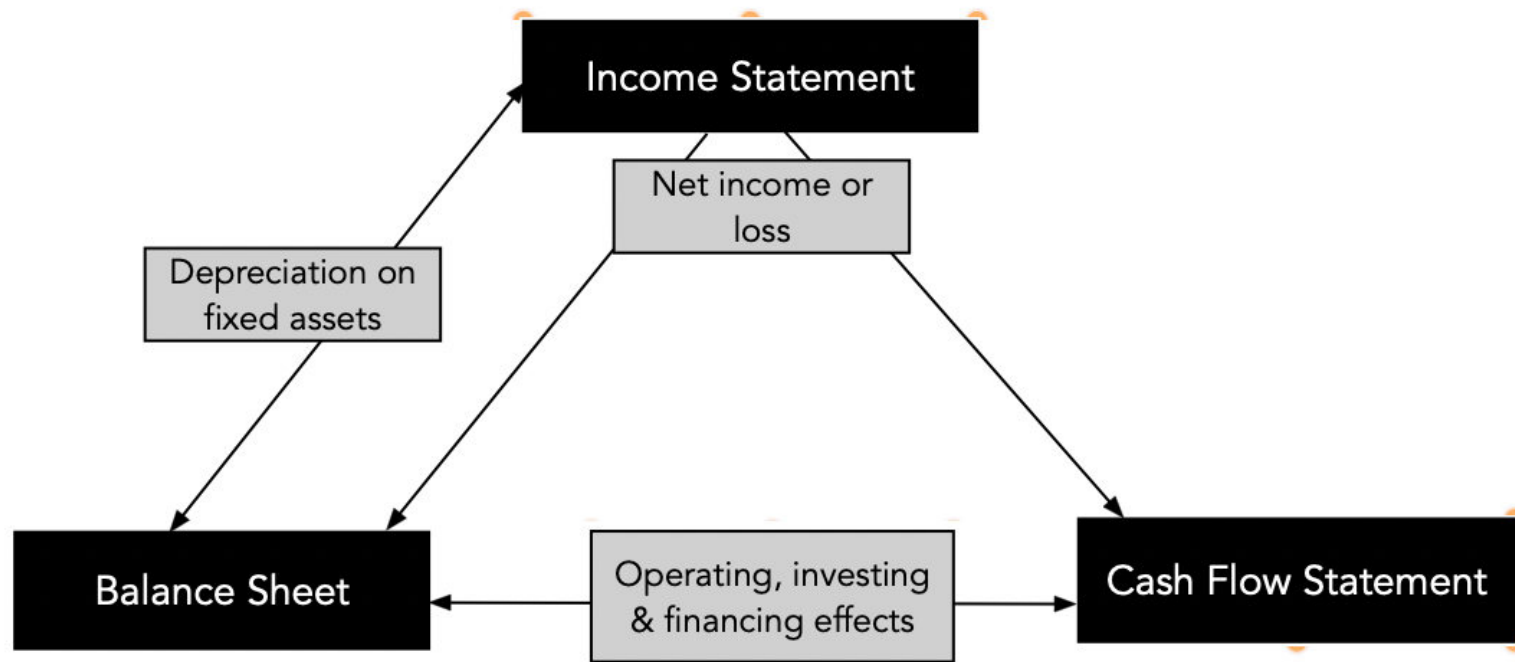
Net cash flow from the issue and repurchase of equity, from the issue and repayment of debt and after dividend payments.

+ Cash Flows from Financing

Net cash flow, after investing and financing

= Net Change in Cash Balance

The Interconnections



The Accounting Standards

- ❑ **Accounting is a rule-driven process**, and over time, those rules have been formalized, especially for publicly traded companies. This formalization is driven by two considerations:
 - ❑ *Standardization*, to allow for comparisons across companies
 - ❑ *First principles*, to ensure that earnings, asset value and cash flows measure what they are supposed to measure.
- ❑ While accounting standards around the world remain different, they have converged (for the most part) around two standards:
 - ❑ **GAAP (Generally Accepted Accounting Principles)**, representing rules developed by FASB (Financial Accounting Standards Board) to cover US financial reporting.
 - ❑ **IFRS (International Financial Reporting Standards)**, representing rules developed by IASB (International Accounting Standards Board) for companies listed globally, followed by about 90 countries as of 2020.



- ☒ IFRS Standards are required for domestic public companies
- ☐ IFRS Standards are permitted but not required for domestic public
- ☐ IFRS Standards are required or permitted for listings by foreign
- ☐ The IFRS for SMEs Standard is required or permitted
- ☐ The IFRS for SMEs Standard is under consideration

The Bottom Line

- The raw material that we use to do financial analysis and valuation almost always takes the form of accounting statements.
- Consequently, it behooves us all to understand how accountants think (even if we disagree with them) in putting these statements together.
- The challenge is that accounting thinking keeps changing, as we move through time, and we have to understand those changes (both the what and the why), to keep up with them.



SESSION 2: INCOME STATEMENTS & PROFITABILITY MEASURES

Accounting for Finance

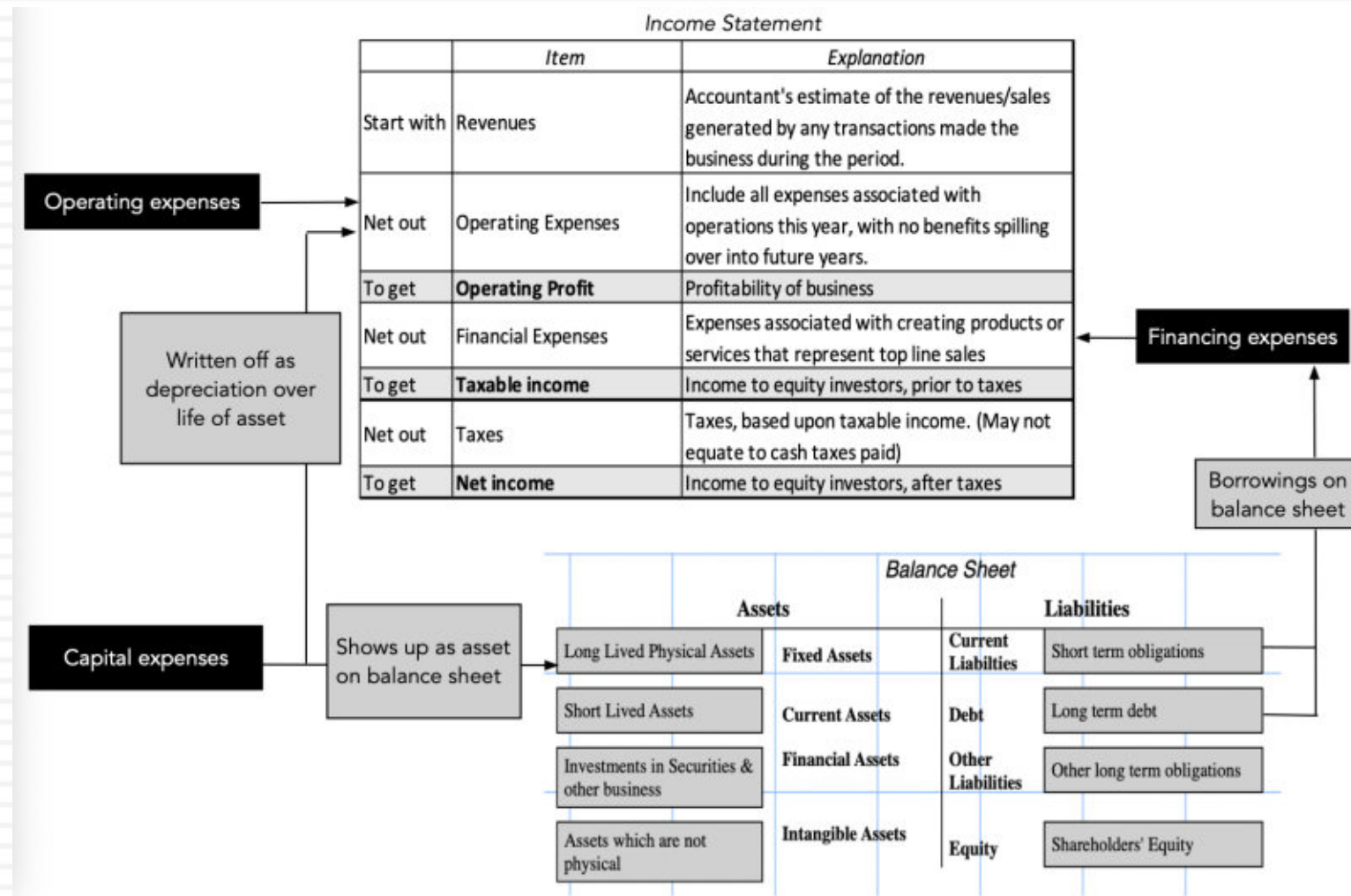
Measuring Income: Accrual versus Cash Accounting

- ❑ In accrual accounting, you record transactions when they occur, rather than when cash flows occur.
 - ❑ Revenues are recorded when a product or service is sold, not when the customer pays for that product or service.
 - ❑ Expenses are recorded consistently, with the expenses associated with producing the sold product or service shown in the period, even though you may have spent the money in a prior period or will not pay until a future period.
- ❑ In cash accounting, you record revenues when you get paid for providing a product or service, and expenses when you pay.
- ❑ Unless you are a small or personal business, you will have to follow accrual accounting rules.

Classifying Expenses: Operating, Financing and Capital Expenses

- **Operating expenses** are expenses associated with the operations of the business. That includes not only the direct costs of producing the product or service the firm sells, but also other expenses associated with production, including S, G & A expenses.
- **Financing expenses** are expenses associated with the use of non-equity financing. Most often, this takes the form of interest expenses on debt.
- **Capital expenses** are expenses that provide benefits over many years. For a manufacturing company, these can take the form of plant and equipment. For non-manufacturing companies, they can take on less conventional and tangible forms (and accounting has never been good at dealing with these).

Everything has a place....



Revisiting the Income Statement

	<i>Item</i>	<i>Explanation</i>
<i>Start with</i>	Revenues	Accountant's estimate of the revenues/sales generated by any transactions made the business during the period.
<i>Net out</i>	Cost of Goods Sold	Expenses associated with producing products or services that represent top line sales
<i>To get</i>	Gross Profit	Production profitability
<i>Net out</i>	Other Operating Expenses	Includes selling, general and administrative and other expenses associated with operations, but not directly tied to producing goods and services
<i>To get</i>	Operating Profit	Profitability of business
<i>Net out</i>	Financial Expenses	Expenses associated with creating products or services that represent top line sales
<i>To get</i>	Taxable Income	Income for equity investors, prior to taxes
<i>Net out</i>	Taxes	Taxes due on taxable income
<i>To get</i>	Net Income	Income for equity investors, after taxes

Revenue Recognition

- For many firms, revenue recognition is a simple process, where once a product or service is sold, it is recorded as revenues.
- For some firms, especially those that sell products or services over many years, it becomes trickier, since the question of how much of the revenue to record in the year of the sale and how much in subsequent years becomes debatable.
 - Under ASC 606: “The new model’s core principle for revenue recognition is to “depict the transfer of promised goods or services to customers in an amount that reflects the consideration to which the entity expects to be entitled in exchange for those goods or services.”
 - Thus, for a real estate developer working on a multi-year construction, revenues should be recognized as construction progresses, and for a software firm that enters in a contract over many years, performance obligations will determine when revenues get recognized.

Revenue Breakdowns

- As companies enter multiple businesses and different geographies, it is useful to know where they generate their revenues.
- While the breakdown can sometimes be provided in income statements, they are more likely to be part of the footnotes to the financial statements:
 - ▣ Companies generally break down *revenues by geography*, though the degree of detail can vary.
 - ▣ Companies also break down *revenues by business segment*, though there is an element of subjectivity to the segment categorization.

Operating Expenses: A Break Down

- COGS versus Other Operating Expenses: Operating expenses are broadly broken down into expenses directly related to producing the goods or services that give rise to revenues, i.e. cost of goods sold, and expenses that are related to operations, but which are not as directly tied to revenues.
 - ▣ The former are netted out from revenues to get to gross profits
 - ▣ The latter get netted out of gross profits to get to operating income
- Selling, General and Administrative Costs: In many companies, the largest non-operating expense is S,G &A, an amorphous item which can include everything but the proverbial kitchen sink.

Depreciation: Accounting, Tax and Economic Forms

- **Economic depreciation** reflects the loss in value (earning power) in an asset, as it ages. It requires nuance, and will vary across even the same type of assets, depending on how it is used.
- **Accounting depreciation** is more mechanical and is driven largely by the aging of the asset, with the differences often being in whether it happens uniformly over the life of the asset or is more accelerated.
- **Tax depreciation** reflects what the tax authorities will allow as depreciation for purposes of computing taxable income.

Financial Expenses

- The most common financial expense is **interest expense on debt**, either in the form of bank loans or corporate bonds.
- As accountants classify other commitments (such as leases) as debt, **some of the interest expense is implicit**, i.e., it is calculated by accountants based upon their assessment of the debt equivalent value of commitments and current interest rates.
- In some companies, interest expenses are netted out against interest income earned by the company on its cash holdings and financial investments, and reported as a **net interest expense**. If interest income exceeds interest expense, this number will measure net interest income.

Income from non-operating investments

- Cash & Marketable Securities: Income earned on cash holdings (which is invested in marketable securities, like treasury bills and commercial paper in most companies) will be reported either as a stand alone income or netted against interest expenses.
- Cross holdings in other companies: The reporting can vary depending upon the magnitude of your holding:
 - When you hold a (small or minority) portion of another company, the income from that holding will usually be reported in the income statement.
 - However, if you hold a majority stake of another company, you will generally have to consolidate your financials. That will require you to count 100% of the subsidiary's revenues, operating expenses and operating income as your own.

Extraordinary Income/ Expenses

- As the term implies, extraordinary income and expenses are designed to capture what a company does not face in the ordinary course of operations.
- Extraordinary items include
 - ▣ One-time expense or gain from sale of assets or divisions
 - ▣ Write offs or charges associated with past project, lawsuits or fines
 - ▣ Impairment of goodwill from acquisitions in the past
- If an item is truly extraordinary, it should show up infrequently and the amount associated with it should vary. If it shows up every year, it is not extraordinary, even if switches signs (goes from profits in some years to losses in others).

Pro-forma Accounting: Con game or legitimate restatement?

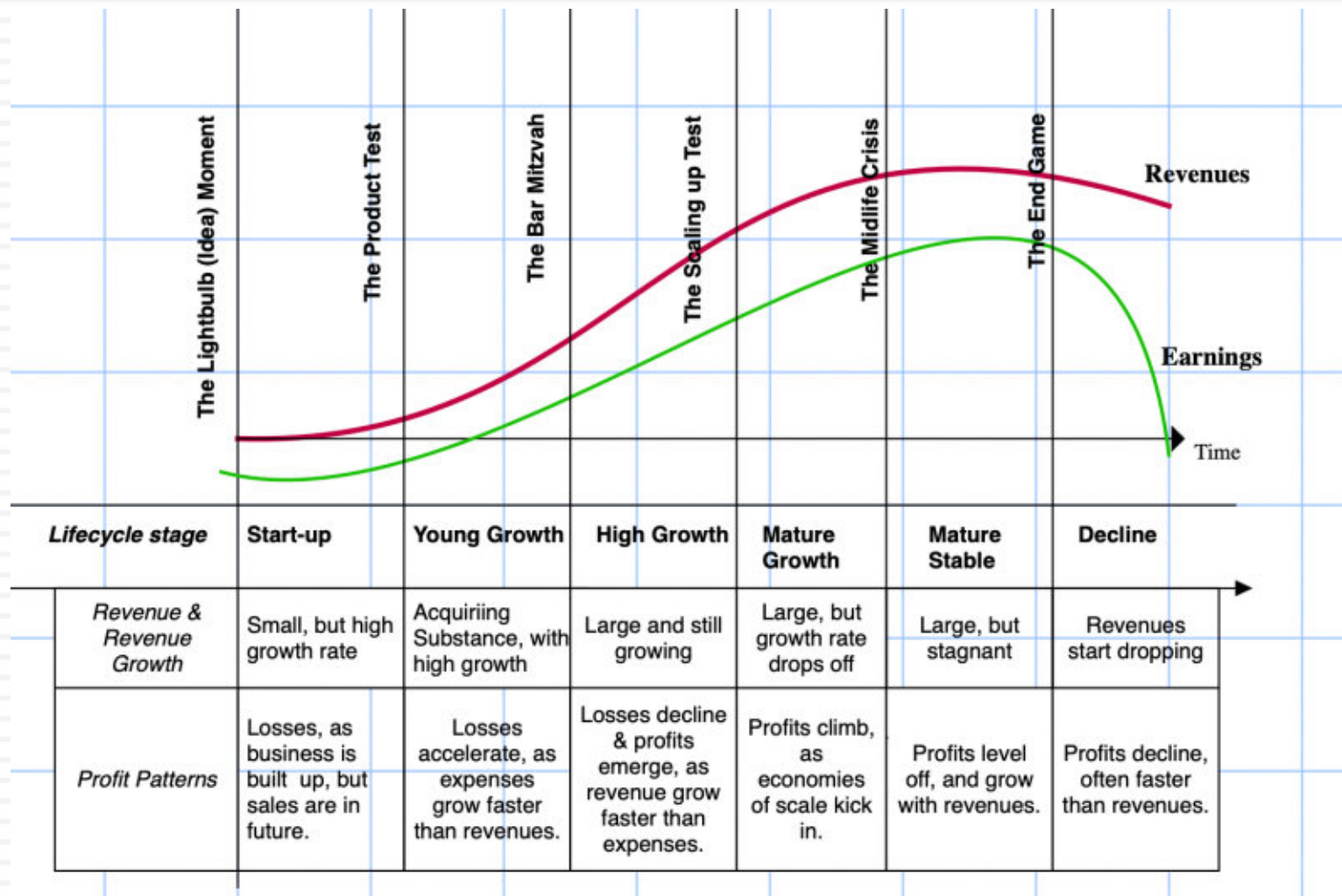
- In recent years, companies have become creative in reporting pro-forma financial statements.
 - In some cases, they do so to correct for what they believe are accounting inconsistencies.
 - In other cases, they are motivated by the desire to increase their profitability.
- As investors, you should never take pro-forma financials at face value but devise your own smell test on what should be added back and what should not to get to proforma income. In general, there are two items you should focus the most attention to:
 - The movement of expenses from operating to capital, sometimes merited, sometimes not.
 - The removal of expenses because they are one time or extraordinary.



SESSION 2A: INCOME STATEMENT COMPARISONS

Accounting for Finance

Income Statement: A Life Cycle Perspective



A Young Company: Peloton's Income Statement (2019) in its Prospectus

PELOTON INTERACTIVE, INC.
CONSOLIDATED STATEMENTS OF OPERATIONS AND COMPREHENSIVE LOSS
(in millions, except share and per share amounts)

	Fiscal Year Ended June 30,		
	2017	2018	2019
Revenue:			
Connected Fitness Products	\$ 183.5	\$ 348.6	\$ 719.2
Subscription	32.5	80.3	181.1
Other	2.6	6.2	14.7
Total revenue	218.6	435.0	915.0
Cost of revenue:			
Connected Fitness Products	113.5	195.0	410.8
Subscription	29.3	45.5	103.7
Other	1.9	4.9	17.0
Total cost of revenue	144.7	245.4	531.4
Gross profit	73.9	189.6	383.6
Operating expenses:			
Research and development	13.0	23.4	54.8
Sales and marketing	86.0	151.4	324.0
General and administrative	45.6	62.4	207.0
Total operating expenses	144.7	237.1	585.8
Loss from operations	(70.7)	(47.5)	(202.3)
Other (expense) income, net:			
Interest (expense) income, net	(0.3)	(0.3)	7.0
Other (expense) income, net	—	—	(0.3)
Total other (expense) income, net	(0.3)	(0.3)	6.7
Loss before provision for income taxes	(71.1)	(47.8)	(195.6)
Provision for income taxes	—	0.1	0.1
Net loss	\$ (71.1)	\$ (47.9)	\$ (195.6)
Net loss attributable to common stockholders	\$ (163.4)	\$ (47.9)	\$ (245.7)
Net loss per share attributable to common stockholders, basic and diluted	\$ (5.97)	\$ (2.18)	\$ (10.72)
Weighted-average common shares outstanding, basic and diluted	27,379,789	21,934,228	22,911,764
Pro forma net loss per share attributable to common stockholders, basic and diluted (unaudited)			\$ (0.84)
Pro forma weighted-average common shares outstanding, basic and diluted (unaudited)			233,552,393
Other comprehensive income:			
Change in unrealized gain (loss) on marketable securities	\$ —	\$ —	\$ 0.2
Total other comprehensive income	—	—	0.2
Comprehensive loss	\$ (71.1)	\$ (47.9)	\$ (195.4)

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A Growth Company: Netflix's Income Statement (2020)

NETFLIX, INC.
CONSOLIDATED STATEMENTS OF OPERATIONS
(in thousands, except per share data)

		Year ended December 31,		
		2019	2018	2017
1	Revenues	\$ 20,156,447	\$ 15,794,341	\$ 11,692,713
	Cost of revenues	12,440,213	9,967,538	8,033,000
	Marketing	2,652,462	2,369,469	1,436,281
	Technology and development	1,545,149	1,221,814	953,710
	General and administrative	914,369	630,294	431,043
	Operating income	2,604,254	1,605,226	838,679
	Other income (expense):			
	Interest expense	(626,023)	(420,493)	(238,204)
	Interest and other income (expense)	84,000	41,725	(115,154)
	Income before income taxes	2,062,231	1,226,458	485,321
	Provision for (benefit from) income taxes	195,315	15,216	(73,608)
	Net income	\$ 1,866,916	\$ 1,211,242	\$ 558,929
	Earnings per share:			
	Basic	\$ 4.26	\$ 2.78	\$ 1.29
	Diluted	\$ 4.13	\$ 2.68	\$ 1.25
	Weighted-average common shares outstanding:			
	Basic	437,799	435,374	431,885
	Diluted	451,765	451,244	446,814

		Year ended December 31,		
		2019	2018	2017
	Net income	\$ 1,866,916	\$ 1,211,242	\$ 558,929
	Other comprehensive income (loss):			
	Foreign currency translation adjustments	(3,939)	975	27,409
	Change in unrealized gains (losses) on available-for-sale securities, net of tax of \$0, \$0, and \$378, respectively	—	—	599
	Total other comprehensive income (loss)	(3,939)	975	28,008
	Comprehensive income	\$ 1,862,977	\$ 1,212,217	\$ 586,937

A Mature Company: Coca Cola (2019)

Year Ended December 31,		2019	2018	2017
1	Net Operating Revenues	\$ 37,266	\$ 34,300	\$ 36,212
	Cost of goods sold	14,619	13,067	13,721
	Gross Profit	22,647	21,233	22,491
2	Selling, general and administrative expenses	12,103	11,002	12,834
	Other operating charges	458	1,079	1,902
	Operating Income	10,086	9,152	7,755
	Interest income	563	689	679
	Interest expense	946	950	853
3	Equity income (loss) — net	1,049	1,008	1,072
	Other income (loss) — net	34	(1,674)	(1,763)
	Income Before Income Taxes	10,786	8,225	6,890
	Income taxes	1,801	1,749	5,607
	Consolidated Net Income	8,985	6,476	1,283
	Less: Net income (loss) attributable to noncontrolling interests	65	42	35
	Net Income Attributable to Shareowners of The Coca-Cola Company	\$ 8,920	\$ 6,434	\$ 1,248
	Basic Net Income Per Share^d	\$ 2.09	\$ 1.51	\$ 0.29
	Diluted Net Income Per Share^d	\$ 2.07	\$ 1.50	\$ 0.29
	Average Shares Outstanding — Basic	4,276	4,259	4,272
	Effect of dilutive securities	38	40	52
	Average Shares Outstanding — Diluted	4,314	4,299	4,324

Revenue Breakdown for Coca Cola: By Geography

	1				2				
	Europe, Middle East & Africa	Latin America	North America	Asia Pacific	Global Ventures	Bottling Investments	Corporate	Eliminations	Consolidated
2019									
Net operating revenues:									
Third party	\$ 6,434	\$ 4,118	\$ 11,906	\$ 4,723	\$ 2,560	\$ 7,431	\$ 94	\$ —	\$ 37,266
Intersegment	624	—	9	604	2	9	—	(1,248)	—
Total net operating revenues	7,058	4,118	11,915	5,327	2,562	7,440	94	(1,248)	37,266
Operating income (loss)	3,551	2,375	2,594	2,282	334	358	(1,408)	—	10,086
Interest income	—	—	65	—	12	—	486	—	563
Interest expense	—	—	—	—	—	—	946	—	946
Depreciation and amortization	86	35	439	31	117	446	211	—	1,365
Equity income (loss) — net	35	(32)	(6)	11	(3)	836	208	—	1,049
Income (loss) before income taxes	3,361	2,288	2,592	2,310	343	716	(824)	—	10,786
Identifiable operating assets	8,143 ¹	1,801	17,687	2,060	7,265	11,170 ¹	18,376	—	66,502
Investments ²	543	716	358	224	14	14,093	3,931	—	19,879
Capital expenditures	108	140	392	47	209	836	322	—	2,054

An Aging Company: Toyota

Consolidated Statements of Income

	(Yen in millions)		
	FY2019 (For the year ended March 31, 2019)	FY2020 (For the year ended March 31, 2020)	Increase (Decrease)
Net revenues:			
Sales of products	28,105,338	27,759,749	(345,589)
Financing operations	2,120,343	2,170,243	49,900
Total net revenues	30,225,681	29,929,992	(295,689)
Costs and expenses:			
Cost of products sold	23,389,495	23,142,744	(246,751)
Cost of financing operations	1,392,290	1,379,620	(12,670)
Selling, general and administrative	2,976,351	2,964,759	(11,592)
Total costs and expenses	27,758,136	27,487,123	(271,013)
Operating income	2,467,545	2,442,869	(24,676)
Other income (expense):			
Interest and dividend income	225,495	232,870	7,375
Interest expense	(28,078)	(32,217)	(4,139)
Foreign exchange gain (loss), net	12,400	(79,020)	(91,420)
Unrealized gains (losses) on equity securities	(341,054)	(24,600)	316,454
Other income (loss), net	(50,843)	14,705	65,548
Total other income (expense)	(182,080)	111,738	293,818
Income before income taxes and equity in earnings of affiliated companies	2,285,465	2,554,607	269,142
Provision for income taxes	659,944	683,430	23,486
Equity in earnings of affiliated companies	360,066	271,152	(88,914)
Net income	1,965,587	2,142,329	176,742
Less – Net income attributable to noncontrolling interests	(102,714)	(66,146)	36,568
Net income attributable to Toyota Motor Corporation	1,862,873	2,076,183	193,310

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Sector and Industry Differences

- The accounting standards that govern how companies measure revenues and earnings may be the same across companies, but there are some sector-specific differences.
 - For commodity companies, the differences often arise from costs expended exploring for commodity reserves and extracting the commodity from reserves.
 - For financial service companies, the challenge often lies in defining revenues and debt, since the latter is less a source of capital and more raw material.

A Commodity Company: Total (France) in 2019

For the year ended December 31, (M\$)(a)		2019	2018	2017
Sales	(Notes 3, 4, 5)	200,316	209,363	171,493
Excise taxes	(Notes 3 & 5)	(24,067)	(25,257)	(22,394)
Revenues from sales	(Notes 3 & 5)	176,249	184,106	149,099
Purchases, net of inventory variation	(Note 5)	(116,221)	(125,816)	(99,411)
Other operating expenses	(Note 5)	(27,255)	(27,484)	(24,966)
Exploration costs	(Note 5)	(785)	(797)	(864)
Depreciation, depletion and impairment of tangible assets and mineral interests	(Note 5)	(15,731)	(13,992)	(16,103)
Other income	(Note 6)	1,163	1,838	3,811
Other expense	(Note 6)	(1,192)	(1,273)	(1,034)
Financial interest on debt		(2,333)	(1,933)	(1,396)
Financial income and expense from cash & cash equivalents		(19)	(188)	(138)
Cost of net debt	(Note 15)	(2,352)	(2,121)	(1,534)
Other financial income	(Note 6)	792	1,120	957
Other financial expense	(Note 6)	(764)	(685)	(642)
Net income (loss) from equity affiliates	(Note 8)	3,406	3,170	2,015
Income taxes	(Note 11)	(5,872)	(6,516)	(3,029)
CONSOLIDATED NET INCOME		11,438	11,550	8,299
Group share		11,267	11,446	8,631
Non-controlling interests		171	104	(332)
Earnings per share (\$)		4.20	4.27	3.36
Fully-diluted earnings per share (\$)		4.17	4.24	3.34

(a) Except for per share amounts.

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A Business Breakdown for Total

1

For the year ended December 31, 2019 (M\$)	Exploration & Production	Integrated Gas, Renewables & Power	Refining & Chemicals	Marketing & Services	Corporate	Intercompany	Total
Non-Group sales	7,201	10,107	07,090	07,200	10	–	200,316
Intersegment sales	31,329	2,825	32,390	659	125	(67,328)	–
Excise taxes	–	–	(3,015)	(21,052)	–	–	(24,067)
REVENUES FROM SALES	38,590	20,992	116,973	66,887	135	(67,328)	176,249
Operating expenses	(16,389)	(18,316)	(112,104)	(63,855)	(925)	67,328	(144,261)
Depreciation, depletion and impairment of tangible assets and mineral interests	(11,659)	(1,492)	(1,527)	(980)	(73)	–	(15,731)
OPERATING INCOME	10,542	1,184	3,342	2,052	(863)	–	16,257
Net income (loss) from equity affiliates and other items	610	2,330	322	101	42	–	3,405
Tax on net operating income	(4,572)	(741)	(470)	(598)	155	–	(6,226)
NET OPERATING INCOME	6,580	2,773	3,194	1,555	(666)	–	13,436
Net cost of net debt							(1,998)
Non-controlling interests							(171)
NET INCOME – GROUP SHARE							11,267

A Financial Service Company: HSBC in 2019

Consolidated income statement for the year ended 31 December

	Notes*	2019 \$m	2018 \$m	2017 \$m
Net interest income		30,462	30,489	28,176
– interest income ^{1,2}		54,695	49,609	40,995
– interest expense ³		(24,233)	(19,120)	(12,819)
Net fee income	2	12,023	12,620	12,811
– fee income		15,439	16,044	15,853
– fee expense		(3,416)	(3,424)	(3,042)
Income/(expense) from trading or managed on a fair value basis	3	10,231	9,531	8,426
Net income/(expense) from assets and liabilities of insurance businesses, including related derivatives, measured at fair value through profit or loss	3	3,478	(1,488)	2,836
Changes in fair value of designated debt and related derivatives ⁴	3	90	(97)	155
Changes in fair value of other financial instruments mandatorily measured at fair value through profit or loss	3	812	695	N/A
Gains less losses from financial investments		335	218	1,150
Net insurance premium income	4	10,636	10,659	9,779
Net operating income		2,957	960	443
Net operating income before change in expected credit losses and other credit impairment charges/Loan impairment charges and other credit risk provisions	4	71,024	63,587	63,776
Change in expected credit losses and other credit impairment charges		(14,926)	(9,807)	(12,331)
Loan impairment charges and other credit risk provisions		56,098	53,780	51,445
Net operating income		(2,756)	(1,767)	N/A
Employee compensation and benefits	5	N/A	N/A	(1,769)
General and administrative expenses		53,342	52,013	49,676
Depreciation and impairment of property, plant and equipment and right-of-use assets ⁵		(18,002)	(17,373)	(17,315)
Amortisation and impairment of intangible assets		(13,828)	(15,353)	(15,707)
Goodwill impairment	21	(2,100)	(1,119)	(1,166)
Operating profit		(1,070)	(814)	(696)
Share of profit in associates and joint ventures	18	(7,349)	—	—
Profit before tax		(42,349)	(34,659)	(34,884)
Tax expense	7	10,993	17,354	14,792
Profit for the year		2,354	2,536	2,375
Attributable to:		13,347	19,890	17,167
– ordinary shareholders of the parent company		(4,639)	(4,865)	(5,288)
– preference shareholders of the parent company		8,708	15,025	11,879
– other equity holders		5,969	12,608	9,683
– non-controlling interests		90	90	90
Profit for the year		1,324	1,029	1,025
Basic earnings per ordinary share	9	1,325	1,298	1,081
Diluted earnings per ordinary share	9	8,708	15,025	11,879
		\$	\$	\$
		0.30	0.63	0.48
		0.30	0.63	0.48

A Pharmaceutical Firm: Dr. Reddy's in 2019

(All amounts in Indian Rupees millions, except share data and where otherwise stated)

Particulars	Note	For the year ended 31 March 2020	For the year ended 31 March 2019
Income			
Sales	2.12	109,925	104,667
Service income and License fees	2.12	8,105	1,062
Other operating income	2.13	474	526
Total revenue from operations		118,504	106,255
Other income	2.14	7,432	2,384
Total income		125,936	108,639
Expenses			
Cost of materials consumed		25,565	21,032
Purchase of stock-in-trade		11,172	8,686
Changes in inventories of finished goods, work-in-progress and stock-in-trade	2.15	(999)	660
Employee benefits expense	2.16	20,302	19,319
Depreciation and amortisation expense	2.17	7,892	7,806
	2.18	478	568
Selling and other expenses	2.19	33,768	33,561
		98,178	91,632
Profit before tax		27,758	17,007
Tax expense/(benefit)	2.26		
Current tax		4,839	2,818
Deferred tax		(6,458)	1,416
Profit for the year		29,377	12,773
Other comprehensive income (OCI)			
Items that will not be reclassified subsequently to profit or loss		88	(1)
Income tax on items that will not be reclassified subsequently to profit or loss		(33)	3
		55	2
Items that will be reclassified subsequently to profit or loss		(750)	209
Income tax on items that will be reclassified subsequently to profit or loss		259	(73)
		(491)	136
Total other comprehensive income/(loss) for the year, net of tax		(436)	138
Total comprehensive income for the year		28,941	12,911
Earnings per share:	2.22		
Basic earnings per share of ₹ 5/- each		177.23	76.98
Diluted earnings per share of ₹ 5/- each		176.88	76.85

The accompanying notes are an integral part of the financial statements.

Bottom Line

- Much is made of the differences in accounting standards around the world and across businesses, but the reality is that accounting standards are converging across the world, rather than diverging.
- While revenues come in different forms for different companies and operating expenses take varied forms, the end game with gross profit, operating and net income remains the same, i.e., to measure profits are different levels.
- In theory, you should be therefore able to compare these numbers across companies, but in practice, accounting inconsistencies in how the expenses get measured and categorized can create problems.



SESSION 3: BALANCE SHEETS - ASSETS OWNED & MONEY OWED

Accounting for Finance

The Balance Sheet: Dueling Views

- ❑ **Record of capital invested:** There are some (including me) who believe that the main function of a balance sheet is to record how much a business has invested in its assets-in-place, i.e., the assets that allow for its current operations to occur.
- ❑ **Measure of current value:** There is a large and perhaps dominant school of thought among accountants, or at least accounting rule writers, that a balance sheet should reflect the value of the business today.
- ❑ **Liquidation value:** There is a third school, with lenders to the firm among its primary members, who feel that a balance should reflect what you would get for the assets of the firm, if you liquidated them today.

Revisiting the Balance Sheet

<i>Balance Sheet</i>			
Assets		Liabilities	
Long Lived Physical Assets	Fixed Assets	Current Liabilities	Short term obligations
Short Lived Assets	Current Assets	Debt	Long term debt
Investments in Securities & other business	Financial Assets	Other Liabilities	Other long term obligations
Assets which are not physical	Intangible Assets	Equity	Shareholders' Equity

Fixed and Current Assets

- ❑ **The Old Way:** If you are old enough to have learned your accounting two or three decades ago, the way you were taught to value fixed and current assets was to show them at original cost, net of accounting depreciation.
- ❑ **The New Way:** As accounting has increasingly adopted the fair value standard, there has been a move to mark assets to current market value.
- ❑ **Divergent Effects:** The difference in values that you get for assets, using the two approaches, varies. It is
 - ❑ Greater on older assets than on newer ones
 - ❑ Greater on fixed assets than current assets

Financial Assets

- Financial assets can take the form of holdings of securities or part ownership of other companies, private or public.
- With holdings of publicly traded securities, the movement to using current market prices to mark up their values is almost complete.
- With equity ownership in other companies, the rules can vary depending on
 - Whether the stake is viewed as a majority stake (>50%) or a minority stake. The former will lead to full consolidation (where 100% of the subsidiaries revenues and operating income will be included in the parent company's financials, with the portion that is not owned shown as minority or non-controlling interest on the liability side) and with the latter, the actual stake will be shown as an asset.
 - With a minority stake, whether it is held for trading or as a long-term investment. With the former, the holding will be marked to market. With the latter, it will be shown at book value terms.

Intangible Assets

- **Big game:** Accountants talk a big game when it comes to intangible assets, and from that talk, you would think that they have figured out how to value the big intangibles (brand name, management quality etc.).
- **But different reality:** In reality, accountants are much better at valuing small-bore intangibles like licenses and customer lists, where the earnings and cash flows from the intangible are observable and forecastable than they are at valuing the big intangibles.

Goodwill: The Most Dangerous Intangible

- After all the talk of intangibles in accounting, it is telling that the bulk of intangible assets on accounting balance sheets across the world take the form of one item: goodwill.
- Goodwill may sound good, but it is a plug variable that signifies little.
 - For goodwill to manifest itself on a balance sheet, a company has to do an acquisition.
 - When that acquisition occurs, goodwill is measured as the difference between the price paid on the acquisition and the target company's asset value (dressed up book value).
 - It shows up as an asset because without it in place, balance sheets would not balance.

Goodwill Impairment: Valuable information or Make-work-for-accountants?

- **Old rules:** For much of the last century, goodwill once created in an acquisition, was written off on autopilot, often amortized over long periods in equal installments.
- **New Rules:** In the late 1990s, both GAAP and IFRS rewrote the rules, requiring accountants to revisit goodwill estimates each year, and make judgments on whether the goodwill had been impaired or not. To make that judgment, accountants would have to revisit the target company valuations and decide whether the value had increased (in which case goodwill would be left unchanged) or decreased (and goodwill would be impaired).
- **Is it informational?** The rationale for this rule change was to provide information to markets, but since goodwill impairments are often based upon market pricing movements (in the sector) and lag them by months and sometimes years, the effect of goodwill impairments on stock prices has been negligible.

Current Liabilities

- ❑ Current liabilities can be broadly broken into three groups:
 - ❑ Non-interest-bearing liabilities, such as accounts payable and supplier credit, which represent part of normal operations.
 - ❑ Interest-bearing short-term borrowings such as commercial paper, short term debt and the short term portion (<1 year) of long term debt.
 - ❑ Deferred salaries, taxes and other amounts due in the short term.
- ❑ When computing non-cash working capital, we do not include interest-bearing short term debt in the calculation, moving it instead into the debt column.

Debt Due

- When companies borrow money, it can take three forms:
 - ▣ *Corporate bonds*, represent debt raised from public markets
 - ▣ *Bank loans*, debt raised from banks and other lending institutions
 - ▣ *Lease debt*, arising out of lease contracts requiring lease payments in future years. Until 2019, only leases classified as capital leases qualified, but since 2019, operating lease commitments are also debt.
- The mark-to-market movement on the asset side of the balance sheet has been muted on the liability side of the balance sheet. Bank debt, for the most part, is recorded as originally borrowed, and corporate bonds due, are for the most part not marked to market.

Debt details

- While balance sheets are the repositories for total debt due, broken down into current and long term, there is additional information on debt in the footnotes, for most companies.
- This additional information can be on two fronts:
 - ▣ Individual debt due, with stated interest rates and maturities.
 - ▣ Additional features on the debt, including floating/fixed and straight/convertible provisions.
 - ▣ A consolidated table of when debt repayments come due, by year.

Shareholder's Equity

- ❑ **Old ways:** The shareholders' equity in a business was a reflection of its entire history, since it started with the equity brought in to start the business, adds on equity augmentations over time as well as the cumulation of retained earnings.
- ❑ **New ways:** The shareholders' equity in a business reflects the jumbled mess of mark-to-market accounting, with all of its contradictions.
- ❑ **My cynical view:** Old or new ways, shareholders' equity (or book equity) has little hope of ever being a measure of the intrinsic value of equity in a business. This quixotic quest on the part of accounting will do more damage than good.

More on shareholders' equity

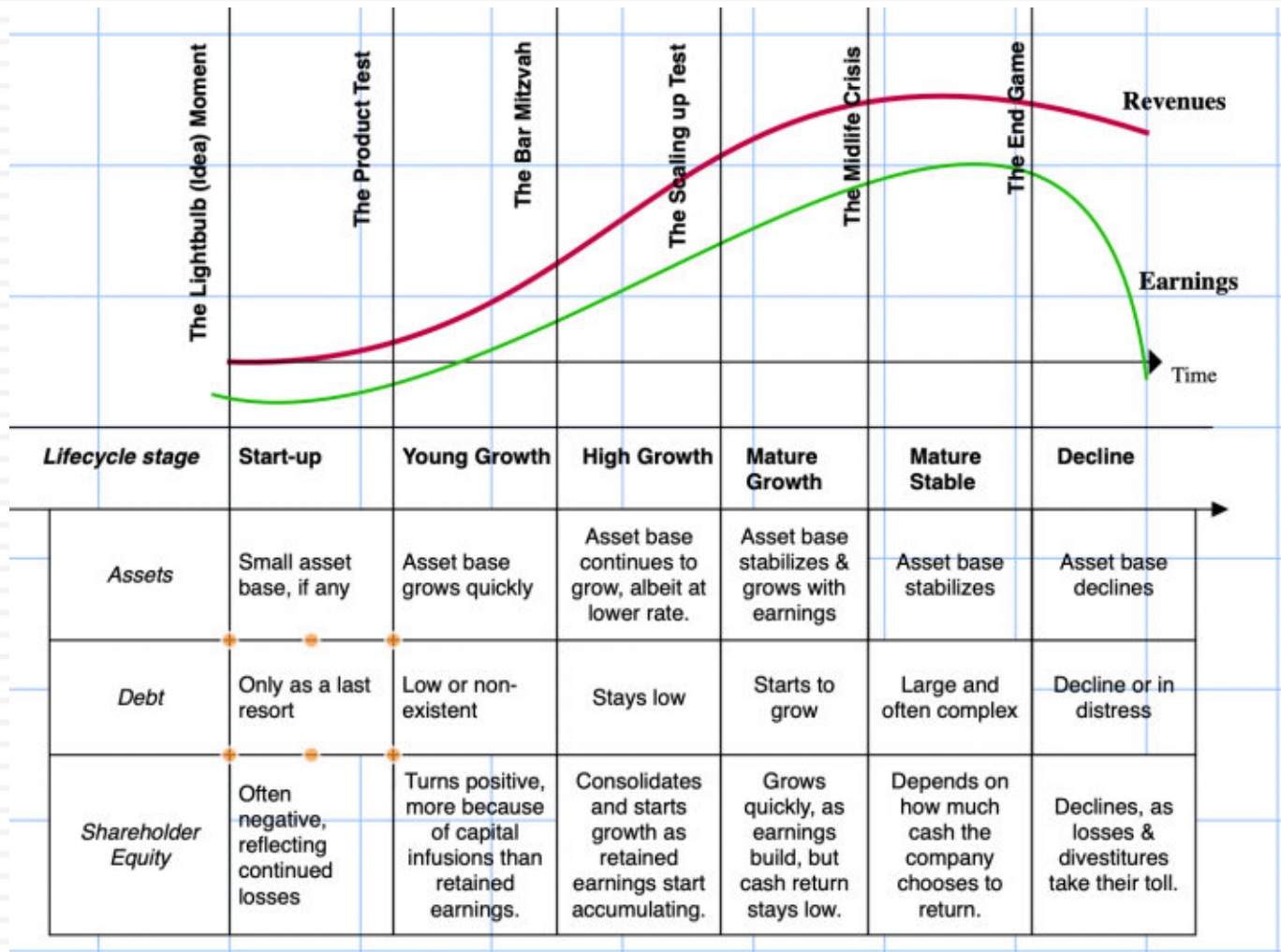
- Par value: This is a throwback in time and should be ignored.
- Company Age: Since shareholders' equity reflects a company's cumulated history of equity raises and retained earnings, young companies will tend to have far less shareholders' equity than older companies, of equivalent market value.
- Capitalization effects: Since only capitalized expenses become part of assets, shareholders' equity can be skewed by accounting rules and corporate actions on what is capitalized and what is expensed.
- Buyback effects: Both dividends and buybacks reduce shareholders' equity, by reducing it, but the magnitude of buybacks makes their effect more dramatic.
- Negative equity? There is no mathematical reason why shareholders' equity cannot become negative, either because a company has lost money for an extended period or because of large buyback/write off.



SESSION 3A: BALANCE SHEET COMPARISONS

Accounting for Finance

Balance Sheet: A Life Cycle Perspective



A Young Company: Peloton's Balance Sheet (2019) in its Prospectus

3

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	As of June 30,		Pro Forma as of June 30, 2019 (unaudited)
	2018	2019	
ASSETS			
Cash and cash equivalents	\$ 150.6	\$ 162.1	
Marketable securities	—	216.0	
Inventories	25.3	136.6	
Prepaid expenses and other current assets	18.4	48.4	
Total current assets	203.8	581.7	
Property and equipment, net	36.2	249.7	
Intangible assets, net	24.5	19.5	
Goodwill	4.2	4.3	
Restricted cash	1.0	0.8	
Other assets	1.6	8.5	
Total assets	\$ 271.2	\$ 864.5	
LIABILITIES, REDEEMABLE CONVERTIBLE PREFERRED STOCK, AND STOCKHOLDERS' DEFICIT			
Current Liabilities:			
Accounts payable	\$ 28.1	\$ 92.2	
Accrued expenses	51.4	104.5	
Customer deposits and deferred revenue	88.5	90.8	
Other current liabilities	2.2	3.3	
Total current liabilities	170.2	290.8	
Deferred rent	9.4	23.7	
Build-to-suit liability	—	147.1	
Other non-current liabilities	1.0	0.4	
Total liabilities	180.5	462.0	
Commitments and contingencies (Note 11)			
Redeemable convertible preferred stock, \$0.000025 par value, 182,193,592 and 215,443,468 shares authorized; 176,313,468 and 210,640,629 shares issued and outstanding as of June 30, 2018 and 2019, respectively; no shares issued and outstanding as of June 30, 2019, pro forma (unaudited)	406.3	941.1	\$ —
Stockholders' Deficit:			
Common stock, \$0.000025 par value, 266,946,216 and 400,000,000 shares authorized; 25,936,848 and 25,301,604 shares issued and outstanding as of June 30, 2018 and 2019, respectively; 235,942,233 shares outstanding pro forma (unaudited)	—	—	—
Additional paid-in capital	20.5	90.7	1,031.8
Accumulated other comprehensive loss	—	0.2	0.2
Accumulated deficit	(315.6)	(538.6)	(629.5)
Total stockholders' deficit	(315.6)	(538.6)	\$ 402.5
Total liabilities, redeemable convertible preferred stock, and stockholders' deficit	\$ 271.2	\$ 864.5	\$ 864.5

A Growth Company: Netflix's Balance Sheet (2019)

		As of December 31,	
		2019	2018
Assets			
Current assets:			
Cash and cash equivalents	\$	5,018,437	\$ 3,794,483
Current content assets, net		—	5,151,186
Other current assets		1,160,067	748,466
Total current assets		6,178,504	9,694,135
Non-current content assets, net		24,504,567	14,951,141
Property and equipment, net		—	—
Other non-current assets		2,727,420	910,843
Total assets	\$	33,975,712	\$ 25,974,400
Liabilities and Stockholders' Equity			
Current liabilities:			
Current content liabilities	\$	4,413,561	\$ 4,681,562
Accounts payable		674,347	562,985
Accrued expenses and other liabilities		843,043	481,874
Deferred revenue		924,745	760,899
Non-current content liabilities		3,334,323	3,759,026
Long-term debt		14,759,260	10,360,058
Other non-current liabilities		1,444,276	129,231
Total liabilities		26,393,555	20,735,635
Commitments and contingencies (Note 5)			
Stockholders' equity:			
Preferred stock, \$0.001 par value; 10,000,000 shares authorized at December 31, 2019 and 2018; no shares issued and outstanding at December 31, 2019 and 2018		—	—
Common stock, \$0.001 par value; 419,000,000 shares authorized at December 31, 2019 and December 31, 2018, respectively; 438,000,049 and 436,598,597 issued and outstanding at December 31, 2019 and December 31, 2018, respectively		2,793,929	2,315,988
Accumulated other comprehensive loss		(23,521)	(19,582)
Retained earnings		4,811,749	2,942,359
Total stockholders' equity		7,582,157	5,238,765
Total liabilities and stockholders' equity	\$	33,975,712	\$ 25,974,400

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A Mature Company: Coca Cola's Balance Sheet (2019)

December 31,	2019	2018
ASSETS		
Current Assets		
Cash and cash equivalents	\$ 6,480	\$ 9,077
Short-term investments	1,467	2,025
Total Cash, Cash Equivalents and Short-Term Investments	7,947	11,102
Marketable securities	3,228	5,013
Trade accounts receivable, less allowances of \$524 and \$501, respectively	3,971	3,685
Inventories	3,379	3,071
Prepaid expenses and other assets	1,886	2,059
Total Current Assets	20,411	24,930
Equity method investments	19,025	19,412
Other investments	604	607
Other assets	6,075	4,148
Deferred income tax assets	2,412	2,674
Property, plant and equipment — net	10,638	9,598
Trademarks with indefinite lives	9,266	6,682
Bottlers' franchise rights with indefinite lives	109	51
Goodwill	16,764	14,109
Other intangible assets	947	743
Total Assets	\$ 86,381	\$ 83,216
LIABILITIES AND EQUITY		
Current Liabilities		
Accounts payable and accrued expenses	\$ 11,312	\$ 9,533
Loans and notes payable	10,994	13,835
Current maturities of long-term debt	4,253	5,003
Accrued income taxes	414	411
Total Current Liabilities	26,973	28,782
Long-term debt	27,516	25,376
Other liabilities	8,510	7,646
Deferred income tax liabilities	2,284	2,354
The Coca-Cola Company Shareowners' Equity		
Common stock, \$0.25 par value; authorized — 11,200 shares; issued — 7,040 shares	1,760	1,760
Capital surplus	17,154	16,520
Reinvested earnings	65,855	63,234
Accumulated other comprehensive income (loss)	(13,544)	(12,814)
Treasury stock, at cost — 2,760 and 2,772 shares, respectively	(52,244)	(51,719)
Equity Attributable to Shareowners of The Coca-Cola Company	18,981	18,981
Equity attributable to noncontrolling interests	2,117	2,077
Total Equity	21,098	19,058
Total Liabilities and Equity	\$ 86,381	\$ 83,216

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Coca Cola's Debt in 2019

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U.S. dollar notes due 2020–2093
U.S. dollar debentures due 2022–2098
U.S. dollar zero coupon notes due 2020
Australian dollar notes due 2020–2024
Euro notes due 2021–2036
Swiss franc notes due 2022–2028
Other, due through 2098 ³
Fair value adjustments ⁴
Total ^{5,6}
Less: Current portion
Long-term debt

December 31, 2019			December 31, 2018		
	Amount	Average Rate ¹		Amount	Average Rate ¹
\$	14,621	2.4 %	\$	13,619	2.6 %
	1,366	4.9		1,390	5.2
	168	8.4		163	8.4
	677	2.4		723	2.2
	12,807	0.5		12,994	0.6
	1,129	3.7		1,128	3.6
	548	6.2		300	4.0
	453	N/A		62	N/A
	31,769	1.9 %		30,379	1.9 %
	4,253			5,003	
\$	27,516		\$	25,376	

2

Maturities of long-term debt for the five years succeeding December 31, 2019 are as follows (in millions):

2020
2021
2022
2023
2024

Maturities of Long-Term Debt
4,253
3,767
3,788
4,097
1,974

Coca Cola's Accounting Intangibles: Small-bore?

Indefinite-Lived Intangible Assets

The following table presents the carrying values of indefinite-lived intangible assets included in our consolidated balance sheets (in millions):

December 31,		2019	2018
Trademarks ¹	\$	9,266	\$ 6,682
Bottlers' franchise rights		109	51
Goodwill ¹		16,764	14,109
Other		110	106
Indefinite-lived intangible assets	\$	26,249	\$ 20,948

Definite-Lived Intangible Assets

The following table provides information related to definite-lived intangible assets (in millions):

	Gross Carrying Value	Accumulated Amortization	Net Carrying Value	Gross Carrying Value	Accumulated Amortization	Net Carrying Value
Customer relationships	\$ 344	\$ (177)	\$ 167	\$ 290	\$ (151)	\$ 139
Bottlers' franchise rights	341	(94)	247	396	(18)	378
Trademarks	177	(99)	78	186	(91)	95
Other	55	(30)	25	88	(61)	27
Total	\$ 917	\$ (400)	\$ 517	\$ 960	\$ (321)	\$ 639

An Aging Company: Toyota's Assets in 2020

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	FY2019 (March 31, 2019)	FY2020 (March 31, 2020)	Increase (Decrease)
Assets			
Current assets:			
Cash and cash equivalents	3,574,704	4,190,518	615,814
Time deposits	1,126,352	828,220	(298,132)
Marketable securities	1,127,160	678,731	(448,429)
Trade accounts and notes receivable, less allowance for doubtful accounts of ¥16,370 million at March 31, 2019 and ¥16,370 million at March 31, 2020	2,372,734	2,094,894	(277,840)
Finance receivables, net	6,647,771	6,614,171	(33,600)
Other receivables	568,156	564,854	(3,302)
Inventories	2,656,396	2,434,918	(221,478)
Prepaid expenses and other current assets	805,964	1,236,225	430,261
Total current assets	18,879,237	18,642,531	(236,706)
Noncurrent finance receivables, net	10,281,118	10,423,858	142,740
Investments and other assets:			
Marketable securities and other securities	7,479,926	7,348,651	(131,275)
Investments			
Affiliated companies	3,313,723	4,123,453	809,730
Employees receivables	21,083	21,484	(199)
Other	1,275,768	1,518,934	243,166
Total investments and other assets	12,091,100	13,012,522	921,422
Property, plant and equipment:			
Land	1,386,308	1,346,988	(39,320)
Buildings	4,802,175	4,730,783	(71,392)
Machinery and equipment	11,857,425	11,939,121	81,696
Vehicles and equipment on operating leases	6,139,163	5,929,233	(209,930)
Construction in progress	651,713	510,963	(140,750)
Total property, plant and equipment, at cost	24,836,784	24,457,088	(379,696)
Less – Accumulated depreciation	(14,151,290)	(13,855,563)	295,727
Total property, plant and equipment, net	10,685,494	10,601,525	(83,969)
Total assets	51,936,949	52,680,436	743,487

An Aging Company: Toyota's Liabilities in 2020

1

	(Yen in millions)		
	FY2019 (March 31, 2019)	FY2020 (March 31, 2020)	Increase (Decrease)
Liabilities			
Current liabilities:			
Short-term borrowings	5,344,973	5,286,026	(58,947)
Current portion of long-term debt	4,254,260	4,574,045	319,785
Accounts payable	2,645,984	2,434,180	(211,804)
Other payables	1,102,802	1,020,270	(82,532)
Accrued expenses	3,222,446	2,926,052	(296,394)
Income taxes payable	320,998	218,117	(102,881)
Other current liabilities	1,335,475	1,443,687	108,212
Total current liabilities	18,226,938	17,902,377	(324,561)
Long-term liabilities:			
Accrued pension and severance costs	963,406	978,626	15,220
Deferred income taxes	1,014,851	1,043,169	28,318
Other long-term liabilities	615,599	821,515	205,916
Total long-term liabilities	13,144,801	13,536,208	391,407
Total liabilities	31,371,739	31,438,585	66,846
Mezzanine equity			
Model AA Class Shares, no par value, authorized: 150,000,000 shares at March 31, 2019 and March 31, 2020 issued: 47,100,000 shares at March 31, 2019 and March 31, 2020	498,073	504,169	6,096
Shareholders' equity			
Toyota Motor Corporation shareholders' equity:			
Common stock, no par value, authorized: 10,000,000,000 shares at March 31, 2019 and March 31, 2020 issued: 3,262,997,492 shares at March 31, 2019 and March 31, 2020	397,050	397,050	—
Additional paid-in capital	487,162	489,334	2,172
Retained earnings	21,987,515	23,427,613	1,440,098
Accumulated other comprehensive income (loss)	(916,650)	(1,166,273)	(249,623)
Treasury stock, at cost, 430,558,325 shares at March 31, 2019 and 496,844,960 shares at March 31, 2020	(2,606,925)	(3,087,106)	(480,181)
Total Toyota Motor Corporation shareholders' equity	19,348,152	20,060,618	712,466
Noncontrolling interests	718,985	677,064	(41,921)
Total shareholders' equity	20,067,137	20,737,682	670,545
Commitments and contingencies			
Total liabilities, mezzanine equity and shareholders' equity	51,936,949	52,680,436	743,487

Sector and Industry Differences

- As with income statements, there are differences in what shows up on balance sheets in different sectors, though the accounting standards governing all companies may be the same.
- In particular, the divergences play out on both sides of the balance sheet:
 - On the asset side, it can show up in how much of the value comes from tangible as opposed to intangible assets.
 - For acquisitive companies, it can also show up as uniquely accounting items like goodwill.

A Commodity Company: Total's Assets in 2020

TOTAL

ASSETS

As of December 31, (M\$)

		2019	2018	2017
Non-current assets				
Intangible assets, net	(Notes 4 & 7)	33,178	28,922	14,587
Property, plant and equipment, net	(Notes 4 & 7)	116,408	113,324	109,397
Equity affiliates: investments and loans	(Note 8)	27,122	23,444	22,103
Other investments	(Note 8)	1,778	1,421	1,727
Non-current financial assets	(Note 15)	912	680	679
Deferred income taxes	(Note 11)	6,216	6,663	5,206
Other non-current assets	(Note 6)	2,415	2,509	3,984
TOTAL NON-CURRENT ASSETS		188,029	176,963	157,683
Current assets				
Inventories, net	(Note 5)	17,132	14,880	16,520
Accounts receivable, net	(Note 5)	13,133	14,255	14,688
Other current assets	(Note 5)	17,013	14,724	14,210
Current financial assets	(Note 15)	3,992	3,654	3,393
Cash and cash equivalents	(Note 15)	27,352	27,907	33,185
Assets classified as held for sale	(Note 2)	1,288	1,364	2,747
TOTAL CURRENT ASSETS		85,265	79,799	84,948
TOTAL ASSETS		273,294	256,762	242,631

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A Commodity Company: Total's Liabilities in 2020

LIABILITIES & SHAREHOLDERS' EQUITY

As of December 31, (M\$)

		2019	2018	2017
Shareholders' equity				
Common shares		8,123	8,227	7,882
Paid-in surplus and retained earnings		121,170	120,569	112,040
Currency translation adjustment		(11,503)	(11,313)	(7,908)
Treasury shares		(1,012)	(1,843)	(458)
TOTAL SHAREHOLDERS' EQUITY – GROUP SHARE	(Note 9)	116,778	115,640	111,556
NON-CONTROLLING INTERESTS		2,527	2,474	2,481
TOTAL SHAREHOLDERS' EQUITY		119,305	118,114	114,037
Non-current liabilities				
Deferred income taxes	(Note 11)	11,858	11,490	10,828
Employee benefits	(Note 10)	3,501	3,363	3,735
Provisions and other non-current liabilities	(Note 12)	20,613	21,432	15,986
Non-current financial debt	(Note 15)	47,773	40,129	41,340
TOTAL NON-CURRENT LIABILITIES		83,745	76,414	71,889
Current liabilities				
Accounts payable		28,394	26,134	26,479
Other creditors and accrued liabilities	(Note 5)	25,749	22,246	17,779
Current borrowings	(Note 15)	14,819	13,306	11,096
Other current financial liabilities	(Note 15)	487	478	245
Liabilities directly associated with the assets classified as held for sale	(Note 2)	795	70	1,106
TOTAL CURRENT LIABILITIES		70,244	62,234	56,705
TOTAL LIABILITIES & SHAREHOLDERS' EQUITY		273,294	256,762	242,631

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A Financial Service Company: HSBC in 2019

1

		At	
	Notes*	31 Dec 2019 \$m	31 Dec 2018 \$m
Items in the course of collection from other banks		154,099	162,843
Hong Kong Government certificates of indebtedness		4,956	5,787
Trading assets	11	38,380	35,859
Financial assets designated and otherwise mandatorily measured at fair value through profit or loss	14	254,271	238,130
Derivatives	15	43,627	41,111
Loans and advances to banks		242,995	207,825
Loans and advances to customers		69,203	72,167
Reverse repurchase agreements – non-trading		1,036,743	981,696
Financial investments	16	240,862	242,804
Prepayments, accrued income and other assets	22	443,312	407,433
		136,680	110,571
		755	684
	18	24,474	22,407
Goodwill and intangible assets	21	20,163	24,357
Deferred tax assets	7	4,632	4,450
Total assets		2,715,152	2,558,124
Liabilities and equity			
Liabilities			
Hong Kong currency notes in circulation		38,380	35,859
Deposits by banks		59,022	56,331
Customer accounts		1,439,115	1,362,643
Repurchase agreements – non-trading		140,344	165,884
Items in the course of transmission to other banks		4,817	5,641
Trading liabilities	23	83,170	84,431
Financial liabilities designated at fair value	24	164,466	148,505
Derivatives	15	239,497	205,835
Debt securities in issue	25	104,555	85,342
Accruals, deferred income and other liabilities	26	118,156	97,380
Current tax liabilities		2,150	718
Liabilities under insurance contracts	4	97,439	87,330
Provisions	27	3,398	2,920
Deferred tax liabilities	7	3,375	2,619
Subordinated liabilities	28	24,600	22,437
Total liabilities		2,522,484	2,363,875
Equity			
Called up share capital	31	10,319	10,180
Share premium account	31	13,959	13,609
Other equity instruments		20,871	22,367
Other reserves		2,127	1,906
Retained earnings		136,679	138,191
Total shareholders' equity		183,955	186,253
Non-controlling interests		8,713	7,996
		192,668	194,249
Total liabilities and equity		2,715,152	2,558,124

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A Pharmaceutical Company: Dr. Reddy's

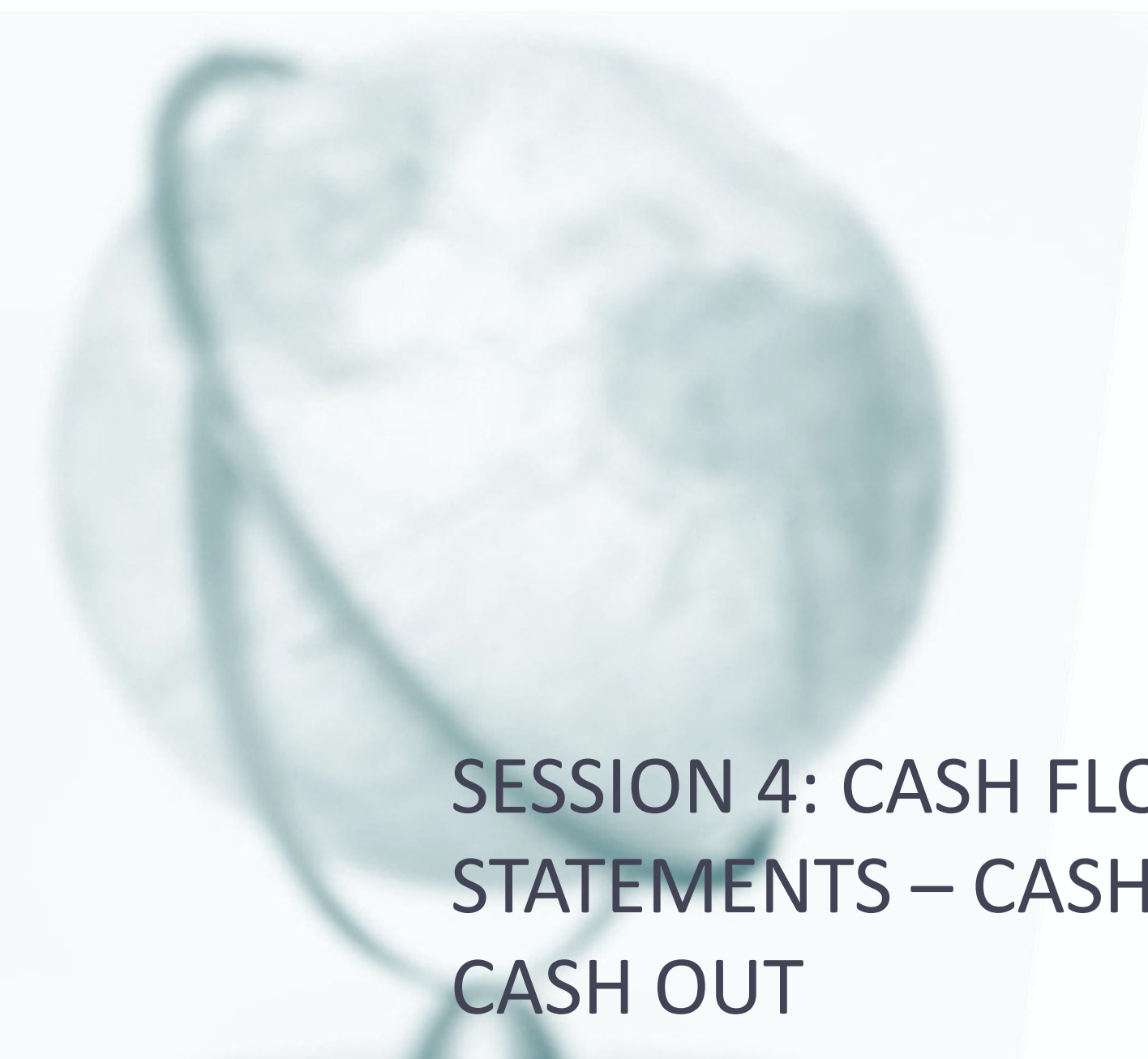
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(All amounts in Indian Rupees millions, except share data and where otherwise stated)

Particulars	Note	As at 31 March 2020	As at 31 March 2019
Assets			
Non-current assets			
Property, plant and equipment	2.1	27,608	20,504
Capital work-in-progress		3,841	4,001
Goodwill	2.2	323	323
Other intangible assets	2.3	6,318	7,000
Intangible assets under development	2.4	277	-
Financial assets			
Investments	2.5 A	33,671	18,151
Trade receivables	2.5 B	1,737	113
Loans	2.5 C	12	332
Other financial assets	2.5 D	474	447
Deferred tax assets, net	2.26	6,129	-
Tax assets, net		3,073	3,106
Other non-current assets	2.6 A	138	126
		93,691	73,143
Current assets			
Inventories	2.7	21,904	20,156
Financial assets			
Investments	2.5 A	21,184	21,144
Trade receivables	2.5 B	46,387	37,177
Derivative instruments	2.29	783	335
Cash and cash equivalents	2.5 E	392	1,132
Other financial assets	2.5 D	1,888	692
Other current assets	2.6 B	8,529	8,696
		101,067	89,332
Total assets		194,758	162,475
Equity and liabilities			
Equity			
Equity share capital	2.8	831	830
Other equity		151,088	126,011
		151,919	126,841
Liabilities			
Non-current liabilities			
Financial liabilities			
Borrowings	2.9 A	193	3,454
Provisions	2.10 A	545	547
Deferred tax liabilities, net	2.26	-	555
Other non-current liabilities	2.11 A	296	285
		1,034	4,841
Current liabilities			
Financial liabilities			
Borrowings	2.9 B	10,436	5,463
Trade payables	2.9 C		
Total outstanding dues of micro enterprises and small enterprises		55	77
Total outstanding dues of creditors other than micro enterprises and small enterprises		10,629	10,239
Derivative instruments	2.29	1,524	45
Other financial liabilities	2.9 D	13,928	10,160
Provisions	2.10 B	2,073	1,847
Other current liabilities	2.11 B	3,160	2,962
		41,805	30,793
Total equity and liabilities		194,758	162,475

Bottom Line

- A balance sheet is the financial statement that most reflects the history of a company, since constructed correctly it is the cumulated result of all of the company's activities during its existence.
- That said, there is disagreement even among accountants as to what the history should reveal, with old-time accounting arguing that it should reflect what the company has invested in its existing assets, not what they are worth today, and fair-value accounting arguing that it should reflect its current value.
- The end result is that balance sheets today are a mess, measuring neither invested capital nor fair value. The most useful items on a balance sheet now are
 - Cash & marketable securities, since it is not subject to nuance
 - Debt, since it measures closely what is owed (at least on interest bearing liabilities)



SESSION 4: CASH FLOW STATEMENTS – CASH IN AND CASH OUT

Accounting for Finance

The End Game with Cash Flows

- ❑ The surface level objective of a statement of cash flows is to explain how much the cash balance of a business changed during a period and why it changed.
- ❑ Embedded in the statement of cash flows, though, is other information including:
 - ❑ How much cash earnings the company had during the period, as contrasted with accrual earnings (in income statements)
 - ❑ How much and where the company reinvested cash during the period to sustain and grow its business
 - ❑ How much cash it raised from or returned to its debt and equity investors
- ❑ The statement of cash flows preserves the signs on cash flows, with negative cash flows shown as minuses and positive cash flows as pluses. It also looks at cash flows through the eyes of equity investors in the company.

Revisiting the Cash Flow Statement

Statement of Cash Flows

Net cash flow from operations, after taxes and interest expenses

Cash Flows From Operations

Includes divestiture and acquisition of real assets (capital expenditures), disposal and purchase of financial assets and cash spent on acquisition of other firms.

+ Cash Flows From Investing

Net cash flow from the issue and repurchase of equity, from the issue and repayment of debt and after dividend payments.

+ Cash Flows from Financing

Net cash flow, after investing and financing

= Net Change in Cash Balance

1. Cash flows from Operations

<i>Cash Flow Effect</i>	<i>Item</i>	<i>Why?</i>
Start with	Net Income	Equity Income
Plus	Depreciation and Amortization	Add back non-cash items
Plus	Other non-cash Expenses	
Plus or Minus	Change in Accounts Receivables	Get to cash to equity from operations
	Change in Inventory	
	Change in Other Current Assets	
	Change in Accounts Payable	
	Change in Taxes Due	
Equals	Cash flow from Operations	

Change in non-cash working capital

The Working Capital Effect?

- Embedded in the cash flow from operations is the change in working capital items, excluding cash
 - ▣ Non-cash Working capital = Non-cash current assets – Non-debt current liabilities
 - ▣ An increase in non-cash working capital will decrease cash flows, whereas a decrease in non-cash working capital will increase cash flows.
- To the extent that non-cash working capital ties up cash and capital, a firm with higher needs for that working capital will have lower cash flows from operations, for any given level of net income, than a firm with lower needs.

2. Cash Flows from Investing

<i>Cash Flow Effect</i>	<i>Item</i>	<i>Why?</i>
Minus	Capital Expenditures	Investment in operating assets
Plus	Divestitures of assets	
Minus	Cash Acquisitions	
Minus	Investments in financial assets	Investment in non-operating assets
Minus	Investments in non-operating assets	
Plus	Divestitures of securities & non-operati	
Equals	Cash flow from Investing	

Operating or Non-operating Assets

- ❑ The investing activities section includes investments in both operating and non-operating assets, except for investment in liquid, close to riskless securities, which is treated as cash & marketable securities.
- ❑ The investments into operating assets, whether internal (cap ex, net of divestitures) or external (acquisitions of other companies) are the engine that drives growth in the operating line items (revenues, operating income etc.) *Note that acquisitions funded with stock will not show up here for obvious reasons.*
- ❑ The investments into non-operating assets create a separate source of value, where the payoff will not show up in the operating line items but below the operating income line, as income from cross holdings or securities.

3. Cash flows from Financing

<i>Action</i>	<i>Item</i>	<i>Why?</i>
Plus	Debt Raised	Net Cash from/to Debt
Minus	Debt Repaid	
Plus	New Equity Issuances	Net Cash from/to Equity Investors
Minus	Dividends Paid	
Minus	Stock Buybacks	
Equals	Cash flow from Financing	

Debt Cash Flows

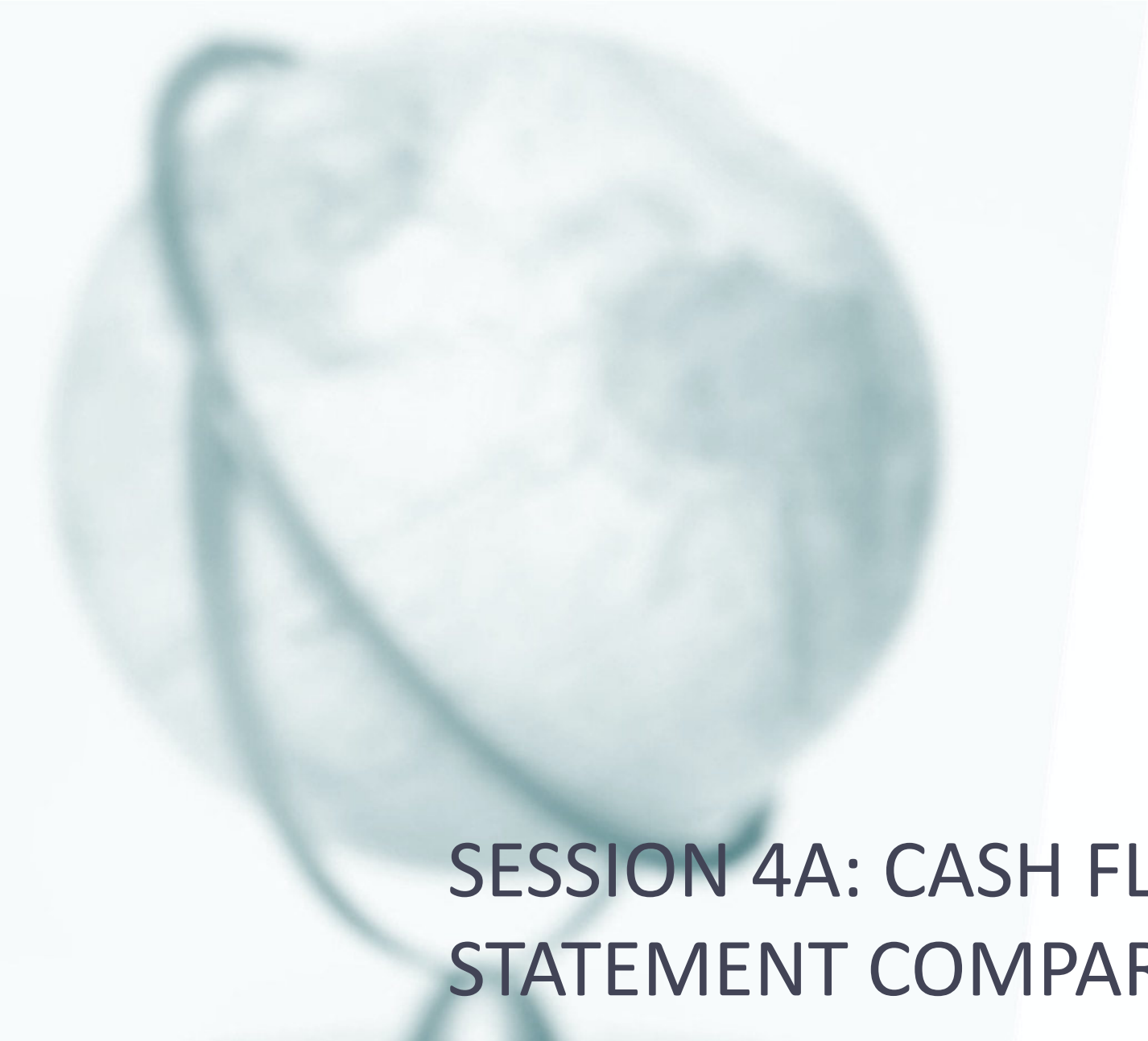
- While interest expenses show up in the operating cash flow section, by reducing net income and showing up in deferred taxes, debt repayments are part of the financing section.
- To the extent that some or all of these debt repayments are funded with debt issuances, the net effect on cash flows can be neutralized or become positive.
- **If total debt increases during a period, it will represent a cash inflow, and if it decreases, it will be a cash outflow. Companies that embark on plans to bring their debt down (up) over time should therefore expect these consequences.**

Dividends and Buybacks

- ❑ Until the 1980s, the only cash flow that was received by equity investors in publicly traded companies was dividends. The effect of paying dividends is simple: it reduces the cash balance at the company and increases the cash in the pockets of every shareholder who receives dividends.
- ❑ Starting in the 1980s, US companies have returned increasing amounts to their shareholders in the form of buybacks.
 - ❑ The effect of buying back stock is exactly the same as paying dividends, to the company, with cash leaving the company.
 - ❑ For shareholders, though, the cash flow effect is disparate. Those shareholders who sell their shares back get cash from the company, and those that do not get no cash, but get a larger share of the equity left in the company.
 - ❑ Both dividends and buybacks reduce shareholder equity on the balance sheet.

Potential Dividends (Free CF to Equity)

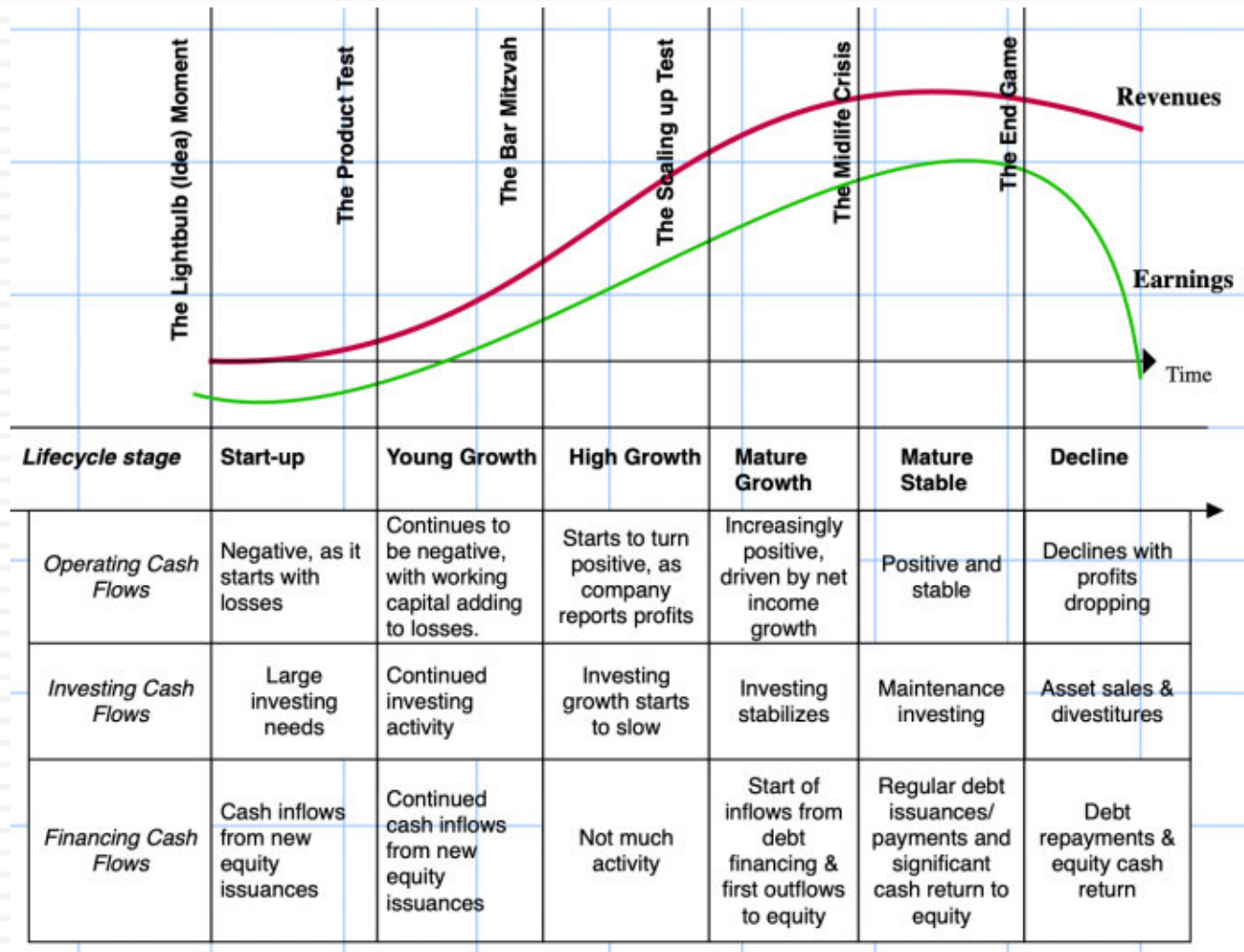
<i>Cash Flow Effect</i>	<i>Item</i>
Start with	Net Income
Plus	Depreciation and Amortization
Plus	Other non-cash Expenses
Plus or Minus	Change in Accounts Receivables
	Change in Inventory
	Change in Other Current Asset
	Change in Accounts Payable
	Change in Taxes Due
Equals	Cash flow from Operations
Minus	Capital Expenditures
Plus	Divestitures of assets
Minus	Cash Acquisitions
Equals	FCFE before debt
Plus	Debt Raised
Minus	Debt Repaid
Equals	FCFE after Debt



SESSION 4A: CASH FLOW STATEMENT COMPARISONS

Accounting for Finance

Cash Flows: A Life Cycle Perspective



A Young Company: Peloton's Cash Flow Statement (2019) in its Prospectus

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	Fiscal Year Ended June 30,		
	2017	2018	2019
Cash Flows from Operating Activities:			
Net loss	\$ (71.1)	\$ (47.9)	\$ (195.6)
Adjustments to reconcile net loss to net cash provided by (used in) operating activities:			
Depreciation and amortization expense	3.7	6.6	21.7
Stock-based compensation expense	10.3	8.5	89.5
Impairment on long-lived assets	0.2	0.7	0.3
Amortization of debt issuance costs	0.1	0.3	0.3
Amortization of (discount) on marketable securities	—	—	(2.2)
Changes in operating assets and liabilities:			
Accounts receivable	(3.6)	(4.1)	(9.1)
Inventories	(5.0)	(9.6)	(111.3)
Prepaid expenses and other current assets	(1.0)	(12.1)	(30.3)
Other assets	(1.1)	1.4	(5.5)
Accounts payable and accrued expenses	22.1	41.0	117.3
Customer deposits and deferred revenue	19.0	63.0	2.2
Other liabilities	3.0	1.0	13.0
Net cash (used in) provided by operating activities	(18.6)	49.7	(108.6)
Cash Flows from Investing Activities:			
Purchases of marketable securities	—	—	(249.8)
Maturities of marketable securities	—	—	36.0
Cash paid for cost method investment	—	—	(0.6)
Purchases of property and equipment	(10.2)	(28.0)	(83.0)
Acquisition of business, net of cash acquired	—	(28.7)	(0.1)
Net cash used in investing activities	(10.2)	(56.7)	(297.5)
Cash Flows from Financing Activities:			
Proceeds from issuance of redeemable convertible preferred stock, net of issuance costs	315.6	—	539.1
Repurchase of common and redeemable convertible preferred stock, including issuance costs	(170.0)	—	(130.3)
Proceeds from borrowings under credit facility	10.5	—	—
Debt repayments	(13.0)	(3.1)	—
Debt issuance costs	(0.2)	(1.2)	(0.9)
Proceeds from exercise of stock options	0.7	7.4	9.3
Net cash provided by financing activities	143.6	3.1	417.2
Effect of exchange rate changes	—	—	0.2
Net change in cash	114.8	(3.9)	11.3
Cash, cash equivalents, and restricted cash — Beginning of period	40.7	155.5	151.6
Cash, cash equivalents, and restricted cash — End of period	\$ 155.5	\$ 151.6	\$ 163.0
Supplemental Disclosures of Cash Flow Information:			
Cash paid for interest	\$ 0.2	\$ 0.3	\$ 1.1
Supplemental Disclosures of Non-Cash Investing and Financing Information:			
Property and equipment accrued but unpaid	\$ 0.4	\$ 4.3	\$ 12.6
Building — build-to-suit asset	\$ —	\$ —	\$ 147.1
Stock-based compensation expense capitalized for software development costs	\$ —	\$ 0.3	\$ 0.8

A Growth Company: Netflix's Cash Flow Statement (2020)

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	Year Ended December 31,		
	2019	2018	2017
Cash flows from operating activities:			
Net income	\$ 1,866,916	\$ 1,211,242	\$ 558,929
Adjustments to reconcile net income to net cash used in operating activities:			
Additions to streaming content assets	(13,916,683)	(13,043,437)	(9,805,763)
Change in streaming content liabilities	(694,011)	999,880	900,006
Amortization of streaming content assets	9,216,247	7,532,088	6,197,817
Depreciation and amortization of property, equipment and intangibles	103,579	83,157	71,911
Stock-based compensation expense	405,376	320,657	182,209
Other non-cash items	228,230	81,640	117,864
Foreign currency remeasurement loss (gain) on long-term debt	(45,576)	(73,953)	140,790
Deferred taxes	(94,443)	(85,520)	(208,688)
Other current assets	(252,113)	(200,192)	(234,090)
Accounts payable	96,063	199,198	74,559
Accrued expenses and other liabilities	157,778	150,422	114,337
Deferred revenue	163,846	142,277	177,974
Other non-current assets and liabilities	(122,531)	2,062	(73,803)
	(2,887,322)	(2,680,479)	(1,785,948)
Cash flows from investing activities:			
Purchases of property and equipment	(253,035)	(173,946)	(173,302)
Change in other assets	(134,029)	(165,174)	(60,409)
Purchases of short-term investments	—	—	(74,819)
Proceeds from sale of short-term investments	—	—	320,154
Proceeds from maturities of short-term investments	—	—	22,705
Net cash provided by (used in) investing activities	(387,064)	(339,120)	34,329
Cash flows from financing activities:			
Proceeds from issuance of debt	4,469,306	3,961,852	3,020,510
Debt issuance costs	(36,134)	(35,871)	(32,153)
Proceeds from issuance of common stock	72,490	124,502	88,378
Other financing activities	—	(1,950)	433
Net cash provided by financing activities	4,505,662	4,048,527	3,076,990
Effect of exchange rate changes on cash, cash equivalents and restricted cash	469	(39,682)	29,848
Net increase in cash, cash equivalents and restricted cash	1,231,745	989,246	1,355,219
Cash, cash equivalents and restricted cash, beginning of year	3,812,041	2,822,795	1,467,576
Cash, cash equivalents and restricted cash, end of year	\$ 5,043,786	\$ 3,812,041	\$ 2,822,795
Supplemental disclosure:			
Income taxes paid	\$ 400,658	\$ 131,069	\$ 113,591
Interest paid	599,132	375,831	213,313

A Mature Company: Coca Cola's Cash Flow Statement

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Year Ended December 31,	2019	2018	2017
Operating Activities			
Consolidated net income	\$ 8,985	\$ 6,476	\$ 1,283
Depreciation and amortization	1,365	1,086	1,260
Stock-based compensation expense	201	225	219
Deferred income taxes	(280)	(913)	(1,252)
Equity (income) loss — net of dividends	(421)	(457)	(628)
Foreign currency adjustments	91	(50)	292
Significant (gains) losses — net	(467)	743	1,459
Other operating charges	127	558	1,218
Other items	504	699	(252)
Net change in operating assets and liabilities	366	(1,240)	3,442
Net Cash Provided by Operating Activities	10,471	7,627	7,041
Investing Activities			
Purchases of investments	(4,704)	(7,789)	(17,296)
Proceeds from disposals of investments	6,973	14,977	16,694
Acquisitions of businesses, equity method investments and nonmarketable securities	(5,542)	(1,263)	(3,809)
Proceeds from disposals of businesses, equity method investments and nonmarketable securities	429	1,362	3,821
Purchases of property, plant and equipment	(2,054)	(1,548)	(1,750)
Proceeds from disposals of property, plant and equipment	978	248	108
Other investing activities	(56)	(60)	(80)
Net Cash Provided by (Used in) Investing Activities	(3,976)	5,927	(2,312)
Financing Activities			
Issuances of debt	23,009	27,605	29,926
Payments of debt	(24,850)	(30,600)	(28,871)
Purchases of stock for treasury	(1,103)	(1,912)	(3,682)
Dividends	(6,845)	(6,644)	(6,320)
Other financing activities	(227)	(272)	(95)
Net Cash Provided by (Used in) Financing Activities	(9,004)	(10,347)	(7,447)
Effect of Exchange Rate Changes on Cash, Cash Equivalents, Restricted Cash and Restricted Cash Equivalents	(72)	(262)	241
Cash, Cash Equivalents, Restricted Cash and Restricted Cash Equivalents			
Net increase (decrease) in cash, cash equivalents, restricted cash and restricted cash equivalents during the year	(2,581)	2,945	(2,477)
Cash, cash equivalents, restricted cash and restricted cash equivalents at beginning of year	9,318	6,373	8,850
Cash, Cash Equivalents, Restricted Cash and Restricted Cash Equivalents at End of Year	6,737	9,318	6,373
Less: Restricted cash and restricted cash equivalents at end of year	257	241	271
Cash and Cash Equivalents at End of Year	\$ 6,480	\$ 9,077	\$ 6,102

An Aging Company: Toyota

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	(Yen in millions)	
	FY2019 (For the year ended March 31, 2019)	FY2020 (For the year ended March 31, 2020)
Cash flows from operating activities:		
Net income	1,985,587	2,142,329
Adjustments to reconcile net income to net cash provided by operating activities		
Depreciation	1,792,375	1,605,383
Provision (reversal) for doubtful accounts and credit losses	80,065	142,982
Pension and severance costs, less payments	31,645	15,699
Losses on disposal of fixed assets	35,902	56,913
Unrealized losses (gains) on securities	339,472	20,082
Deferred income taxes	(86,594)	192,147
Equity in earnings of affiliated companies	(500,000)	(271,132)
Changes in operating assets and liabilities, and other	(51,789)	(313,740)
Net cash provided by operating activities	3,766,597	3,590,643
Cash flows from investing activities:		
Additions to finance receivables	(15,884,610)	(16,896,588)
Collection of and proceeds from sales of finance receivables	14,859,103	15,667,462
Additions to fixed assets excluding equipment leased to others	(1,452,725)	(1,407,832)
Additions to equipment leased to others	(2,286,162)	(2,187,299)
Proceeds from sales of fixed assets excluding equipment leased to others	65,437	48,751
Proceeds from sales of equipment leased to others	1,385,074	1,391,215
Purchases of marketable securities and security investments	(1,840,355)	(2,413,087)
Proceeds from sales of and maturity of marketable securities and security investments	2,698,798	2,370,406
Payment for additional investments in affiliated companies, net of cash acquired	5,010	(14,763)
Changes in investments and other assets, and other	(246,811)	290,874
Net cash used in investing activities	(2,697,241)	(3,150,861)
Cash flows from financing activities:		
Proceeds from issuance of long-term debt	5,000,921	5,691,499
Payments of long-term debt	(4,442,232)	(4,424,923)
Dividends paid to Toyota Motor Corporation class shareholders	(8,690)	(11,186)
Dividends paid to Toyota Motor Corporation common shareholders	(636,116)	(618,801)
Dividends paid to noncontrolling interests	(69,367)	(54,945)
Net cash provided by (used in) financing activities	(540,839)	397,138
Effect of exchange rate changes on cash and cash equivalents and restricted cash and cash equivalents	(41,641)	(131,245)
Net increase in cash and cash equivalents and restricted cash and cash equivalents	486,876	705,675
Cash and cash equivalents and restricted cash and cash equivalents at beginning of year	3,219,639	3,706,515
Cash and cash equivalents and restricted cash and cash equivalents at end of year	3,706,515	4,412,190

Sector and Industry Differences

- The statement of cash flows is perhaps the most direct of financial statements, since it is based upon cash flows in and cash flows out. It is thus the one statement that accounting game playing affects the least.
- While the objective of the statement might be explaining why the cash balance changes, it is the repository for all of the items needed to estimate cash flows.
- While it is not uncommon for income statements and cash flows to send divergent messages about a firm's financial well being, consistent differences between the two can be a indicator of accounting problems.

A Commodity Company: Total in 2019

TOTAL

For the year ended December 31, (M\$)

CASH FLOW FROM OPERATING ACTIVITIES

		2019	2018	2017
Depreciation, depletion, amortization and impairment	(Note 5.3)	16,401	14,584	16,611
Non-current liabilities, valuation allowances, and deferred taxes	(Note 5.3)	(50)	(507)	(534)
(Gains) losses on disposals of assets		(614)	(930)	(2,598)
Undistributed affiliates' equity earnings		(1,083)	(826)	42
(Increase) decrease in working capital	(Note 5.5)	(1,718)	769	827
Other changes, net		319	443	(478)

CASH FLOW FROM OPERATING ACTIVITIES

CASH FLOW USED IN INVESTING ACTIVITIES

Intangible assets and property, plant and equipment additions	(Note 7)	(11,810)	(17,080)	(13,767)
Acquisitions of subsidiaries, net of cash acquired		(4,748)	(3,379)	(800)
Investments in equity affiliates and other securities		(1,618)	(1,108)	(1,368)
Increase in non-current loans		(1,061)	(618)	(961)

Proceeds from disposals of intangible assets and property, plant and equipment		527	3,716	1,036
Proceeds from disposals of subsidiaries, net of cash sold		158	12	2,909
Proceeds from disposals of non-current investments		349	1,444	294

Total divestments

		2,060	7,239	5,264
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CASH FLOW USED IN INVESTING ACTIVITIES

		(17,177)	(14,946)	(11,632)
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CASH FLOW FROM FINANCING ACTIVITIES

Issuance (repayment) of shares:

– Parent company shareholders		452	498	519
– Treasury shares		(2,810)	(4,328)	–

Dividends paid:

– Parent company shareholders		(6,641)	(4,913)	(2,643)
– Non-controlling interests		(115)	(97)	(141)

Net issuance of perpetual subordinated notes	(Note 9)	–	–	–
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Payments on perpetual subordinated notes	(Note 9)	(371)	(325)	(276)
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Other transactions with non-controlling interests		10	(622)	(4)
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Net issuance (repayment) of non-current debt	(Note 15)	8,131	649	2,277
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Increase (decrease) in current borrowings		(5,829)	(3,990)	(7,175)
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Increase (decrease) in current financial assets and liabilities		(536)	(797)	1,903
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CASH FLOW FROM / (USED IN) FINANCING ACTIVITIES

		(7,709)	(13,925)	(5,540)
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NET INCREASE (DECREASE) IN CASH AND CASH EQUIVALENTS

		(201)	(4,168)	5,147
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Effect of exchange rates		(354)	(1,110)	3,441
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Cash and cash equivalents at the beginning of the period		27,907	33,185	24,597
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CASH AND CASH EQUIVALENTS AT THE END OF THE PERIOD	(Note 15)	27,352	27,907	33,185
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A Financial Service Company: HSBC in 2019

Consolidated statement of cash flows for the year ended 31 December

	2019 \$m	2018 \$m	2017 \$m
Profit before tax	13,347	19,890	17,167
Adjustments for non-cash items:			
Depreciation, amortisation and impairment ¹	10,519	1,933	1,862
Net gain from investing activities	(399)	(126)	(1,152)
Share of profits in associates and joint ventures	(2,354)	(2,536)	(2,375)
Gain on disposal of subsidiaries, businesses, associates and joint ventures	(929)	—	(79)
Change in expected credit losses gross of recoveries and other credit impairment charges	3,012	2,280	N/A
Loan impairment losses gross of recoveries and other credit risk provisions	N/A	N/A	2,603
Provisions including pensions	2,423	1,944	917
Share-based payment expense	478	450	500
	(2,297)	(1,303)	(381)
	(3,742)	4,930	(20,757)
Change in net trading securities and derivatives	(18,910)	20,855	(13,615)
Change in loans and advances to banks and customers	(53,760)	(44,071)	(108,984)
Change in reverse repurchase agreements – non-trading	(7,390)	(25,399)	(37,281)
Change in financial assets designated and otherwise mandatorily measured at fair value	(2,308)	(1,515)	(5,303)
Change in other assets	(21,863)	6,766	(6,570)
Change in deposits by banks and customer accounts	79,163	(5,745)	102,457
Change in repurchase agreements – non-trading	(25,540)	35,882	41,044
Change in debt securities in issue	19,268	18,806	(1,369)
Change in financial liabilities designated at fair value	20,068	4,500	8,508
Change in other liabilities	23,124	(2,187)	13,514
Dividends received from associates	633	910	740
Contributions paid to defined benefit plans	(533)	(332)	(685)
Tax paid	(2,267)	(3,417)	(3,175)
Net cash from operating activities	29,743	32,515	(12,414)
	(445,907)	(399,458)	(357,264)
	413,186	386,056	418,352
Net cash flows from the purchase and sale of property, plant and equipment	(1,343)	(1,196)	(1,167)
Net cash flows from purchase/(disposal) of customer and loan portfolios	1,118	(204)	6,756
Net investment in intangible assets	(2,289)	(1,848)	(1,285)
Net cash flow on disposal of subsidiaries, businesses, associates and joint ventures	(83)	4	165
Net cash from investing activities	(35,318)	(16,646)	65,557
Issue of ordinary share capital and other equity instruments	—	6,001	5,196
Cancellation of shares	(1,000)	(1,998)	(3,000)
Net sales/(purchases) of own shares for market-making and investment purposes	141	133	(67)
Redemption of preference shares and other equity instruments	—	(6,078)	—
Subordinated loan capital repaid ³	(4,210)	(4,077)	(3,574)
Dividends paid to shareholders of the parent company and non-controlling interests	(9,773)	(10,762)	(9,005)
Net cash from financing activities	(14,842)	(16,781)	(10,450)
Net increase/(decrease) in cash and cash equivalents	(20,417)	(912)	42,693
Cash and cash equivalents at 1 Jan ⁴	312,911	323,718	263,324
Exchange differences in respect of cash and cash equivalents	1,248	(9,895)	17,701
Cash and cash equivalents at 31 Dec^{4, 5}	293,742	312,911	323,718
Cash and cash equivalents comprise:			
– cash and balances at central banks	154,099	162,843	180,624
– items in the course of collection from other banks	4,956	5,787	6,628
– loans and advances to banks of one month or less	41,626	39,460	61,973
– reverse repurchase agreements with banks of one month or less	65,370	74,702	58,850
– treasury bills, other bills and certificates of deposit less than three months	20,132	21,685	11,593
– cash collateral and net settlement accounts	12,376	14,075	10,900
– less: items in the course of transmission to other banks	(4,817)	(5,641)	(6,850)
Cash and cash equivalents at 31 Dec^{4, 5}	293,742	312,911	323,718

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A Pharmaceutical Firm: Dr. Reddy's Lab in 2019

STATEMENT OF CASH FLOWS

(All amounts in Indian Rupees millions, except share data and where otherwise stated)

Particulars	For the year ended 31 March 2020	For the year ended 31 March 2019
Cash flows from/(used in) operating activities		
Profit before tax	27,758	17,007
Adjustments:		
Depreciation and amortisation expense	7,892	7,806
Impairment loss on other intangible assets	-	24
Equity settled share-based payment expense	521	389
Fair value gain on financial instruments at fair value through profit or loss	(821)	(669)
Foreign exchange loss / (gain), net	(229)	2,455
(Profit)/Loss on sale/disposal of property, plant and equipment and other intangible assets, net	135	(400)
Interest income	(856)	(812)
Finance costs	478	568
Allowances for credit losses, net	93	212
Allowances for doubtful advances, net	2	(351)
Dividend from subsidiary, joint ventures and other entities	(397)	-
Provision/(reversal of provision) relating to non-current investments	-	359
Changes in operating assets and liabilities:		
Trade receivables	(10,927)	4,547
Inventories	(1,748)	(1,588)
Trade payables	368	(201)
Other assets and other liabilities, net	892	663
Cash generated from operations	23,161	30,009
Income taxes paid, net	(4,769)	(2,388)
Net cash from operating activities	18,392	27,621
Cash flows from/(used in) investing activities		
Proceeds from sale of property, plant and equipment	58	879
Expenditures on property, plant and equipment	(4,262)	(5,775)
Expenditures on other intangible assets	(476)	(753)
Purchase of investments	(122,726)	(77,267)
Proceeds from sale of investments	109,186	74,786
Loans and advances repaid by subsidiaries	343	1,800
Dividend income received	397	-
Interest income received	588	821
Net cash used in investing activities	(16,892)	(5,509)
Proceeds from issuance of equity shares (including treasury shares)	4	-
Proceeds from/(repayment of) short-term loans and borrowings, net (Refer note 2.9 (d))	4,630	(17,049)
Repayment of long-term loans and borrowings, net (Refer note 2.9 (d))	(1,805)	-
Payment of principal portion of lease liabilities (Refer note 2.9 (d))	(155)	-
Dividends paid (including corporate dividend tax)	(3,914)	(4,002)
Purchases of treasury shares	(474)	(535)
Interest paid	(527)	(645)
Net cash used in financing activities	(2,241)	(22,231)
Net increase / (decrease) in cash and cash equivalents	(741)	(119)
Effect of exchange rate changes on cash and cash equivalents	-	44
Cash and cash equivalents at the beginning of the year (Refer note 2.5 E)	1,132	1,207
Cash and cash equivalents at the end of the year (Refer note 2.5 E)	391	1,132

Bottom Line

- The statement of cash flows looks at cash flows through the eyes of the equity investors in a business.
 - ▣ Thus, if you are trying to compute cash flows to equity investors, all of the information you will need should be on the statement.
 - ▣ You should also be able to see what equity investors received as cash flows.
- If your objective is to estimate cash flows, prior to debt payments, you can start with the statement of cash flows but you have to trace interest expenses (netted out to get to net income) and add them back.



SESSION 5: CLEANING UP ACCOUNTING

Accounting for Finance

The Accountant's Role

- ❑ Accountants like order and consistency, as can be seen in their propensity to write rules.
- ❑ That said, much of accounting as practiced today was developed in detail in the 20th century for the manufacturing firms that dominated that century.
- ❑ As the center of economic gravity has shifted from manufacturing to technology & service companies, and corporate financial behavior has changed over time, accountants have struggled with four key issues:
 - ❑ Taxes, and the actions that companies take to avoid or delay paying them.
 - ❑ Managerial compensation in the form of equity (stock)
 - ❑ Commitments that are contractual but are not debt
 - ❑ Investments for long term benefits that are not in physical assets

1. Taxes: Dueling Tax Rates

1. Marginal tax rate: The marginal tax rate is the tax rate in the statutory tax codes. Thus, in 2020, a US company should be paying 21% of its taxable income in the US as federal taxes. Since US companies now operate on a regional tax model, the marginal tax rate for multinationals will reflect where they make (or report to make) their taxable income.
2. Effective Tax Rate: The effective tax rate for a company reflects the taxes and taxable income it reports in its income statement, which is based on accrual accounting:
 - ▣ $\text{Effective tax rate} = \text{Taxes} / \text{Taxable Income}$
3. Cash Tax Rate: The cash tax rate for a company reflects the taxes it actually pays on its taxable income.

Deferred Tax Assets & Liabilities

- ❑ For most companies, the effective tax rate will be lower than the marginal tax rate, reflecting:
 - ❑ Operations in countries with lower tax rates
 - ❑ (Legal) Tax deferral and avoidance strategies
- ❑ When there are differences between what is expensed and what is reported as taxable income between the reporting and tax books, the resulting difference in taxes is reported as a deferred tax liability (asset) if the company pays less (more) in taxes on its tax books than it reports in its financial statements.
- ❑ The logic for doing so is simple. The items that give rise to less (more) taxes paid in the current period will reverse and result in more (less) taxes paid in future periods.

Net Operating Losses & Carryforwards

- When a US company loses money, it is allowed to carry those losses forward and use them to reduce taxes paid in future years.
 - ▣ Until the Tax Reform Act of 2017, the NOL could be carried back two years and used to reduce taxes paid in prior years (as a tax credit) and forward 20 years.
 - ▣ The Tax Reform Act of 2017 removed the carry back provision and allows losses to be carried forward indefinitely.
- When a company has losses that are over multiple years, these losses are cumulated over time as a Net Operating Loss (NOL) and should be disclosed in a company's financials.

2. Stock Based Compensation is an Operating Expense...

- ❑ Companies have used stock-based compensation to reward employees for decades for two reasons:
 - ❑ To align the interests of employees & managers with those of the shareholders
 - ❑ To make up for the absence of cash (to provide compensation packages that are competitive)
- ❑ Stock-based compensation primarily takes two forms:
 - ❑ Options to buy the company's stock (or invest in its equity) at a fixed price for a specified period.
 - ❑ Shares in the company, sometimes with restrictions on trading on those shares (restricted shares)

What type of expense is it?

- No matter what the motive for providing stock-based compensation (to align interests or to make up for lack of cash), it is clearly a compensation expense.
 - If a grant is large and occasional, and primarily driven by the desire to align interests, there is an argument that it should be spread out over time.
 - If a company uses stock-based compensation consistently, and more to make up for its cash poor status than to align interests, it is an annual expense.
- To expense stock-based compensation, you have to value of the options or stock given to employees, at the time they are granted.
 - Until 2004, companies were allowed leeway to estimate the value of option grants based upon exercise value at the time of the grant.
 - After 2004, FAS 123 requires companies to value options based upon their time premium (using option pricing models) and show that expense in the year the options are granted.

Is it a cash flow?

- ❑ Now that options and restricted stock are treated (correctly) as compensation, and expensed in the years they are granted, the debate has shifted to the question of whether they are non-cash expenses (like depreciation), deserving of being added back to get to cash flows.
- ❑ Most companies and analysts seem to have come down on yes as the answer, and many companies add back stock-based compensation to get to adjusted earnings.
- ❑ I disagree strongly. There is a fundamental difference between a non-cash expense like depreciation, where you pay nothing, and giving a share of equity (options or shares) in lieu of cash.

3. Leases are debt

- The Essence of Debt: When you borrow money, you create contractual obligations for the future, and a failure to meet them can put your survival as a going concern at risk.
- Lease contracts: When you sign a lease contract, you create commitments for the future, and a failure to meet these commitments will put your survival at risk. Put simply, there is no reason (and there never has been) to treat leases as debt.
- Accounting for leases: Until 2019, accountants disagreed and broke leases down into two groups:
 - Capital leases, where the lessee has effective ownership of the asset, is treated as debt, with a counter asset.
 - Operating leases, where the lessee has (temporary) use of the asset for a period, were treated as operating expenses, with no debt or counter assets on the balance sheet.

Starting in 2019, both GAAP and IFRS are requiring companies to treat all lease commitments as debt, no matter how structured.

Capitalizing Leases

- The process of converting lease commitments to debt follows a simple process, akin to how any bank debt or corporate bond can be valued.
- Here are the steps:
 - ▣ Start with the contractual lease commitments for future years, by year.
 - ▣ Compute the pre-tax cost of borrowing for the firm today, based upon its default risk.
 - ▣ Take the present value of lease commitments, using the pre-tax cost of debt as your discount rate.
- The present value of lease commitments is treated as debt, with the same value shown as a counter-asset.
- To complete the cycle, you compute interest expenses on the lease debt and depreciation on the counter asset and bring them into your income statement.
 - ▣ Interest expense on lease debt = PV of lease commitments * Pre-tax cost of debt
 - ▣ Depreciation on lease asset is computed using the life of the lease (as the life of the asset) and the depreciation method chosen.

Other Contractual Commitments

- While accounting has (finally) come to terms with treating leases as debt, there are a whole host of contractual commitments that share the same characteristics as leases, and require the same treatment.
- Here are some examples:
 1. *Purchase commitments* for many manufacturing firms
 2. *Content commitments* at a streaming company (like Netflix)
 3. *Player contracts* for a sports team

4. R&D is a cap ex

- If the essence of a capital expenditure is that it is an expense whose expected benefits are not just in the current period, but in future periods, research and development (R&D) expenses clearly fit the bill.
- That said, accounting rules around the world, for the most part, require companies to expense R&D, using one of two rationale:
 - ▣ The benefits of R&D are too uncertain. Consequently, they should be expensed until the R&D is closer to commercial development.
 - ▣ It is better to be conservative in estimating earnings.
- Neither justification makes sense.
 - ▣ Uncertainty is never used with other types of capital expenditure (building a factory to make a new and untested product) as the divining rod for capital vs operating expenses.
 - ▣ Accounting should deliver the most realistic estimate of earnings, not the most conservative.

Capitalizing R&D

- To capitalize R&D, there are three steps:
 - Step 1: Estimate an amortizable life for R&D by making your best judgment on how long it takes, on average, for R&D to pay off (as commercial success).
 - Step 2: Collect R&D expenses from past years (going back as long as the amortizable life). If your company has not been in existence for that long, collect as many years as you can.
 - Step 3: For each of the past years of R&D, estimate
 - How much you will be amortizing this year
 - How much of the R&D expense remains unamortized
- To complete the cycle, here are the last steps:
 - Adjust earnings by adding back the current year's R&D expense and subtracting out the amortization of past years R&D
 - Show the unamortized R&D as an asset, and show the same amount as an increase in book equity.

Other Capital Investments

- There are other expenses that fit the R&D profile, i.e., expenses designed to create benefits over many years, but since these investments are not in physical assets, they are treated as operating expenses.
- Here are some examples:
 - ▣ **Advertising expenses** by a consumer product company to build up brand name
 - ▣ **Recruiting and training expenses** by a consulting firm to build its consulting practice
 - ▣ **Exploration costs** for an oil company
 - ▣ **Customer acquisition costs** for a subscriber or user based company.

The Bottom Line: Trust, but verify...

- Accounting statements reflect not only an “accounting” view of the company, but the burden of accounting history and legacy rules.
- When analyzing a company, you should start with accounting statements, but you should have no qualms about changing, modifying or redoing them to reflect what you are trying to do with the data in those statements.



SESSION 5A: ACCOUNTING INCONSISTENCY EXAMPLES

Accounting for Finance

1. Tax Rates

	<i>Peloton</i>	<i>Netflix</i>	<i>Coca Cola</i>	<i>Toyota</i>	<i>HSBC</i>	<i>Dr. Reddy's</i>
<i>In Income Statement</i>						
Taxable Income	\$ (196)	\$ 2,062	\$ 10,786	¥2,285,465	HK\$13,347	₹ 27,758
Taxes	\$ -	\$ 195	\$ 1,801	¥659,944	HK\$4,639	-₹ 1,619
Effective Tax Rate	0.00%	9.47%	16.70%	28.88%	34.76%	-5.83%
<i>In cash flow statement</i>						
Deferred taxes	\$ -	\$ (94)	\$ (421)	-¥86,694	HK\$0	₹ 0
Taxes paid					-HK\$2,267	-₹ 4,769
<i>In balance sheet</i>						
Deferred Tax Assets (Liabilities)	\$ 89	\$ 658	\$ (2,284)	-¥1,014,851	-HK\$3,375	₹ 6,129

2. Non-debt Commitments

- ❑ In general, interest-bearing debt will show up on balance sheets, though some of it may be included in current liabilities (if due in less than a year) and the rest as debt.
- ❑ There are other contractual commitments that have historically not shown up on balance sheets, but should be treated like debt.
- ❑ The most common of these commitments is operating leases, but IFRS and GAAP have finally made the 'right' decision and started including them as debt in 2019.

A Retail Example: Nordstrom's

Fiscal year	Operating Leases
2020	\$333
2021	353
2022	327
2023	300
2024	252
Thereafter	1,136
Total lease payments	
Less: amount representing interest	
Present value of net lease payments ¹	

Discounted at
4.7%

1

On the balance sheet in 2019

Current liabilities:		
Accounts payable	\$1,576	\$1,469
Accrued salaries, wages and related benefits	510	580
Current portion of operating lease liabilities	244	—
Long-term debt, net		
Deferred property incentives, net	4	457
Non-current operating lease liabilities	1,875	—
Other liabilities	683	498

2

Consequences for the company

- **Debt increases:** The debt on the balance sheet is augmented by the present value of lease commitments.
- **A counter asset is created:** Equivalent to the lease debt.
- **Operating income changes,** since you add back the current year's lease expense and reduce it by depreciation. Net income does not or should not change.
 - ▣ *Interest expenses go up* by the interest portion of the current year's lease payment.
 - ▣ *Depreciation is increased* by the depreciation on the lease asset (using the prior year's value)
- *Taxes and net income do not change,* since you replace one tax deductible expense (operating lease) with two (interest and depreciation) of equivalent value.

Another example: Netflix

1

	As of December 31,	
	2019	2018
	(in thousands)	
Less than one year	\$ 8,477,367	\$ 8,611,398
Due after one year and through 3 years	8,352,731	8,841,561
Due after 3 years and through 5 years	2,041,340	1,684,582
Due after 5 years	618,644	148,334
Total streaming content obligations	<u>\$ 19,490,082</u>	<u>\$ 19,285,875</u>

2

	2019	2018
Assets		
Current assets:		
Cash and cash equivalents	\$ 5,018,437	\$ 3,794,483
Current content assets, net	—	5,151,186
Other current assets	1,160,067	748,466
Total current assets	6,178,504	9,694,135
Non-current content assets, net	24,504,567	14,951,141
Property and equipment, net	565,221	418,281
Other non-current assets	2,727,420	910,843
Total assets	<u>\$ 33,975,712</u>	<u>\$ 25,974,400</u>
Liabilities and Stockholders' Equity		
Current liabilities:		
Current content liabilities	\$ 4,413,561	\$ 4,681,562
Accounts payable	674,347	562,985
Accrued expenses and other liabilities	843,043	481,874
Deferred revenue	924,745	760,899
Total current liabilities	6,855,696	6,487,320
Non-current content liabilities	3,334,323	3,759,026
Long-term debt	14,759,260	10,360,058
Other non-current liabilities	1,444,276	129,231
Total liabilities	<u>26,393,555</u>	<u>20,735,635</u>

3. Non-Physical Capital Expenses

- While accountants almost always treat investments in physical assets as capital expenses and show them on the balance sheet, they are inconsistent and unpredictable when it comes to investments in non-physical assets.
 - A pharmaceutical company that buys a patent from another one is allowed to treat that expenditure as a capital expenditure, but one that does R&D to arrive at the same result is not.
 - A company reliant on human capital for its value is almost never required to treat what it invests in human capital for the long term (recruiting and training, for example) as capital expenditures.
 - A subscriber/user based company that spends money acquiring users or subscribers generally is not allowed to treat the money spent acquiring customers as capital expenditures.

A Pharmaceutical Company Example

Step 1: Determining an amortizable life for R & D expenses⁽¹⁾

How long will it take, on an expected basis, for research to pay off at Amgen? Given the length of the approval process for new drugs by the Food and Drugs Administration, we will assume that this amortizable life is 10 years.

Step 2: Capitalize historical R&D expense

Year	R&D Expense	Unamortized portion	Amortization this year
Current	3030.00	1.00 3030.00	
-1	3266.00	0.90 2939.40	\$326.60
-2	3366.00	0.80 2692.80	\$336.60
-3	2314.00	0.70 1619.80	\$231.40
-4	2028.00	0.60 1216.80	\$202.80
-5	1655.00	0.50 827.50	\$165.50
-6	1117.00	0.40 446.80	\$111.70
-7	864.00	0.30 259.20	\$86.40
-8	845.00	0.20 169.00	\$84.50
-9	823.00	0.10 82.30	\$82.30
-10	663.00	0.00 0.00	\$66.30
		\$13283.60	\$1,694.10

Current year's R&D expense = Cap ex = \$3,030 million
 R&D amortization = Depreciation = \$ 1,694 million
 Unamortized R&D = Capital invested (R&D) = \$13,284 million

Step 3: Restate earnings, book value and return numbers

	Unadjusted	Adjusted for R&D	Comments
Net Income	\$4,196	$4,196 + 3030 - 1694 = \$ 5,532$	Add current year's R&D and subtract R&D amortization
Book value of equity	\$17,869	$17,869 + 13,284 = \$ 31,153$	Add unamortized R&D from prior years
Return on Equity	$\frac{4196}{17869} = 23.48\%$	$\frac{5532}{31153} = 17.75\%$	Return on equity drops when book equity is augmented by R&D, even though net income rises.
Pre-tax Operating Income	\$5,594	$5,594 + 3030 - 1694 = \$ 6,930$	Add current year's R&D and subtract R&D amortization
Book value of invested capital	\$21,985	$\$21,985 + \$13,284 = \$35,269$	Add unamortized R&D from prior years
Pre-tax Return on Capital	$\frac{5594}{21985} = 25.44\%$	$\frac{6930}{35269} = 19.65\%$	Return on capital drops when capital is augmented by R&D, even though operating income rises.

4. Stock Based Compensation

- Companies following IFRS and GAAP report the current year's stock-based compensation as an operating expense, valuing both options and restricted stock at the time of issuance.
 - ▣ That said, the residue of past option grants will show up in the footnotes of these companies, with relevant information on remaining maturity and exercise price.
 - ▣ The restricted stock units granted in past years will show up as part of the discussion of share count.
- In both cases, companies will then try to reverse the accounting charge, claiming it is non-cash in reporting pro-forma or adjusted earnings.

Options Outstanding... at Netflix

Stock Option Plans

In June 2011, the Company adopted the 2011 Stock Plan. The 2011 Stock Plan provides for the grant of incentive stock options to employees and for the grant of non-statutory stock options, stock appreciation rights, restricted stock and restricted stock units to employees, directors and consultants.

A summary of the activities related to the Company's stock option plans is as follows:

	Options Outstanding		Weighted- Average Exercise Price (per Share)	Weighted- Average Remaining Contractual Term (in Years)	Aggregate Intrinsic Value (in Thousands)
	Shares Available for Grant	Number of Shares			
Balances as of December 31, 2016	13,289,953	22,437,347	\$ 44.83		
Granted	(2,550,038)	2,550,038	159.56		
Exercised	—	(3,338,474)	26.79		
Expired	—	(1,561)	3.25		
Balances as of December 31, 2017	10,739,915	21,647,350	\$ 61.13		
Granted	(2,039,974)	2,039,974	311.66		
Exercised	—	(3,205,911)	38.66		
Expired	—	(2,135)	\$ 4.60		
Balances as of December 31, 2018	8,699,941	20,479,278	\$ 89.61		
Granted	(2,588,380)	2,588,380	320.66		
Exercised	—	(2,208,052)	32.88		
Expired	—	(280)	6.74		
Balances as of December 31, 2019	6,111,561	20,859,326	\$ 124.28	5.57	\$ 4,230,013
Vested and exercisable at December 31, 2019		20,859,326	\$ 124.28	5.57	\$ 4,230,013

Stock-based compensation expense

405,376

And Adjusted EBITDA... at Peloton

The following table presents a reconciliation of Adjusted EBITDA to net loss, the most directly comparable financial measure prepared in accordance with GAAP, for each of the periods indicated:

	Fiscal Year Ended June 30,		
	2017	2018	2019
	(dollars in millions)		
Net loss(1)	\$ (71.1)	\$ (47.9)	\$ (195.6)
Adjusted to exclude the following:			
Other (expense) income, net	(0.3)	(0.3)	6.7
Provision for income taxes	—	0.1	0.1
Depreciation and amortization expense	2.7	6.6	21.7
Stock-based compensation expense	10.3	8.5	89.5
Transaction costs	—	0.5	0.4
Litigation expenses	5.0	1.5	12.1
Ground lease expense related to build-to-suit obligations	—	—	7.2
Adjusted EBITDA	\$ (51.8)	\$ (30.4)	\$ (71.3)
Adjusted EBITDA Margin	(23.7)%	(7.0)%	(7.8)%

1

Bottom Line

- Accounting statements are just raw material: In corporate finance and valuation, financial statements are raw data that should be viewed as accounting opinion and not fact.
- That you should mold to your own needs: To the extent that accounting perspectives can be outdated or reflect a different set of priorities, you should feel no qualms about redoing or reconstructing accounting statements.
- But your choices will have consequences: That can have consequences for how you measure profits, invested capital and even share count. The company that you see will be closer to the truth than the company described in accounting statements.
- Markets don't follow (and are more sensible than) accountants: For the most part, markets learn and move on faster than accountants do. Thus, markets have been pricing in retailers, on the assumption that leases are debt for decades, while accountants made the change in 2019.



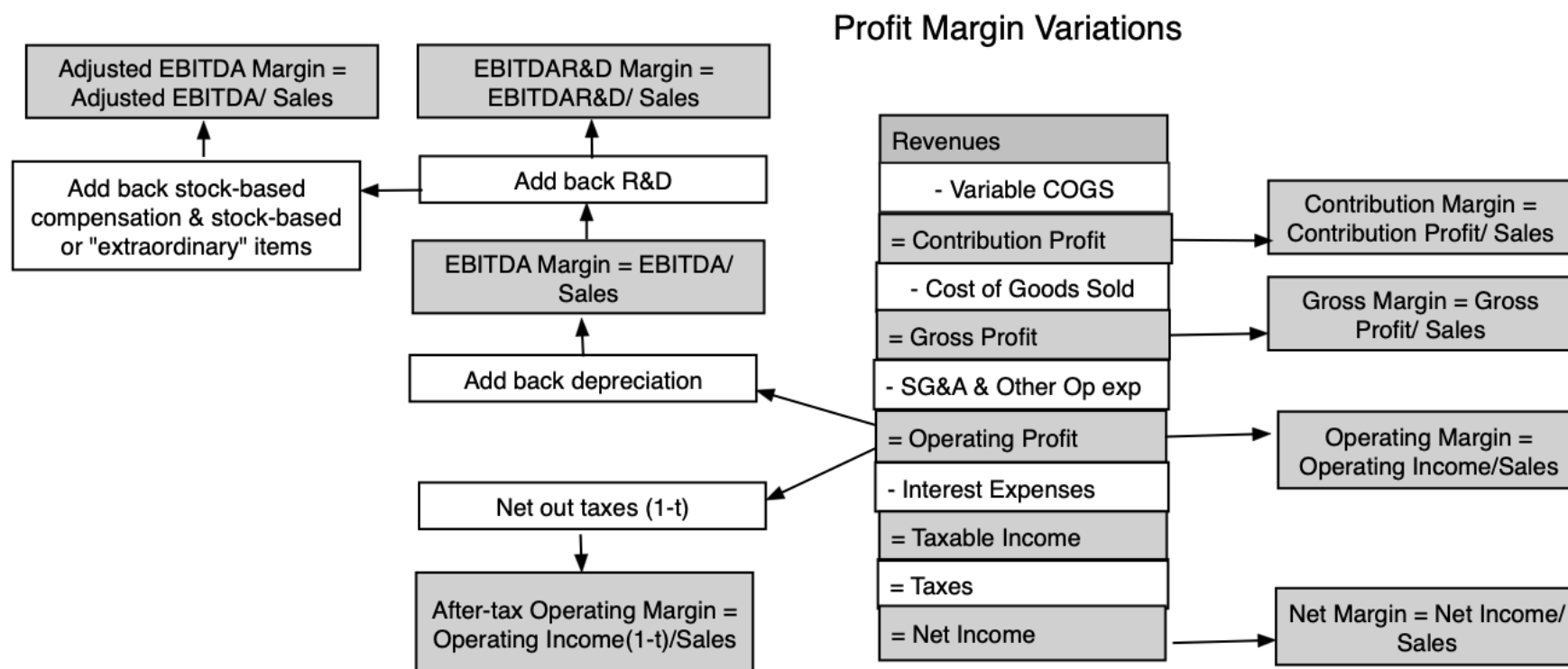
SESSION 6: ACCOUNTING FINANCIAL RATIOS – PROFITABILITY MEASURES

Accounting for Finance

From Absolutes to Ratios

- ❑ Financial statements measure operations in absolute terms, i.e., in dollars, rupees or reals, depending upon the currency of denomination.
- ❑ Absolute measures are difficult, if not impossible, to compare across companies, since bigger companies, all else held constant, should have higher dollar profits and carry more debt.
- ❑ Ratios scale absolute values to each other, and allow for:
 - ❑ Comparisons across companies
 - ❑ Comparisons across time
 - ❑ Comparison to benchmarks

1. Profit Margins



Contribution & Gross Margins: The Costs of Production

- *Contribution margin* measures the pure profits that you generate with every marginal unit you sell, since it nets out only the variable cost associated with producing that unit, giving many software companies close to 100% contribution margins.
- *Gross margins* are a close relative, providing a direct measure of marginal profitability and an indirect measure of how revenue increases flow into profits. To illustrate, Zoom, one of the few stocks that has seen its value increase during the crisis, reported a gross margin of 92% in 2019.
- **Companies with high contribution and high gross margins have much more profit potential, other things remaining equal, than companies with low margins.**

Operating Margins: Measures and Implications

- *Operating margins* measure what is left after the other operating expenses of the company, which cannot be directly traced to individual unit sales, but are nevertheless necessary for its operations.
- To the extent that these other operating costs (like SG&A) are fixed (or more fixed) than the costs of production, the difference between gross and operating margins becomes a simple proxy for potential economies of scale.
- **Companies with high gross margins and low operating margins should see operating profits (and margins) improve much faster as they scale up than companies where operating and gross margins are similar.**

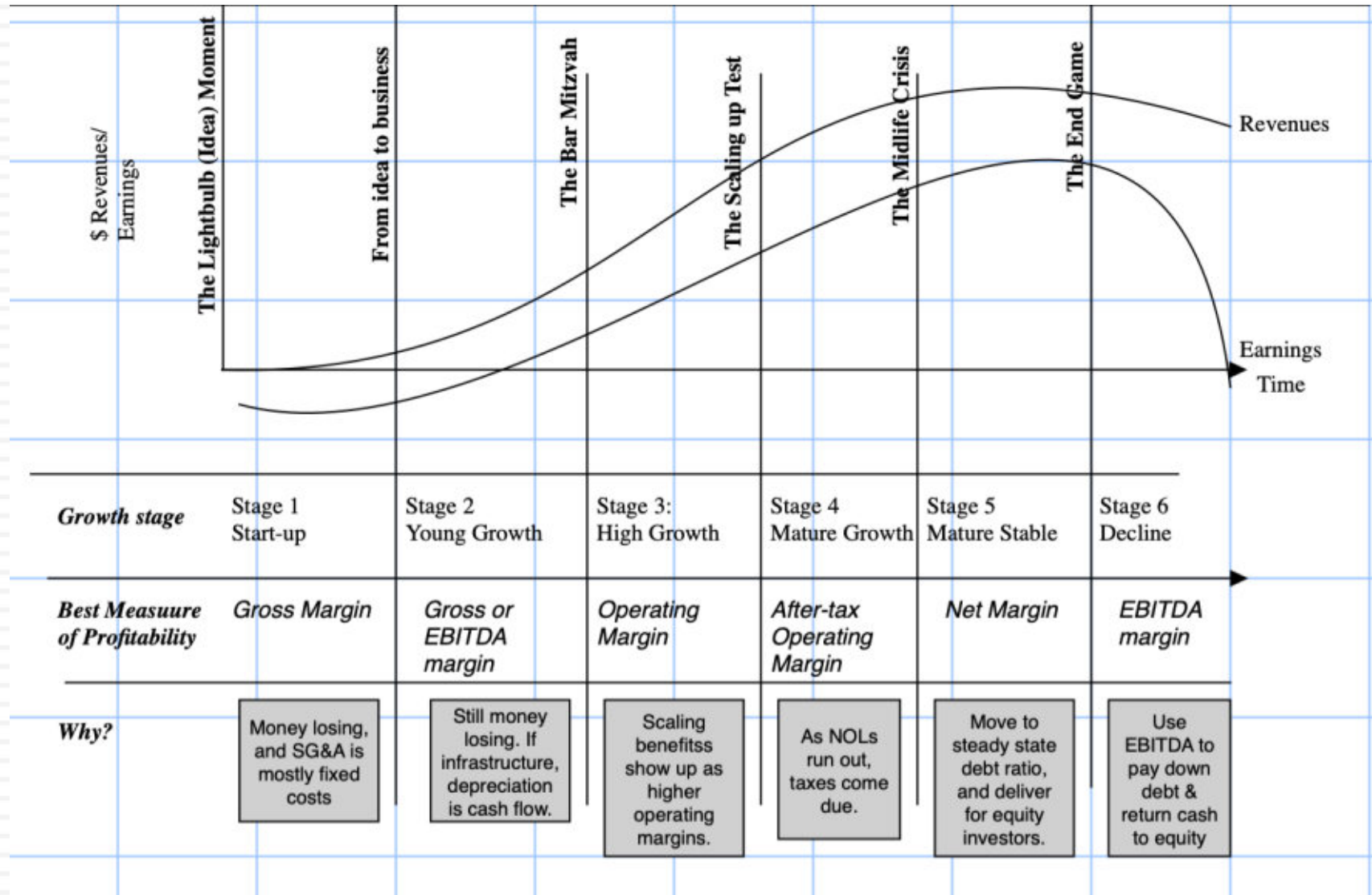
EBITDA Margin: Measures and Implications

- The EBITDA is a rough measure of operating cash flows, rough because it is before taxes and capital expenditures.
- Notwithstanding that, it remains a measure of the cash generating capacity of a company, prior to discretionary choices (on how much to reinvest and borrow) and is used by
 - ▣ Lenders to determine whether the company can afford to borrow money, since debt has to be paid before capital expenditures are made.
 - ▣ Equity investors to decide whether the entire business is fairly valued, before it tries to expand its asset base.
- **Companies with high EBITDA margins generate higher cash flows per dollar of revenues and should be able to borrow more than companies with lower EBITDA margins.**

Net Margins: Measures and Implications

- Netting out taxes and interest expenses, and adding back income from cash and cross holdings, yields *net margin*, a measure of what equity investors get to keep out of every dollar of revenues.
- It is a mixed and noisy measure, reflecting a company's operating model, its tax liabilities and its financial leverage (since debt creates interest expenses and affects taxes), as well as non-operating assets.
- **Companies with high net margins deliver more profits for equity investors, in the aggregate, but perhaps not a per share basis (if debt is the reason why net margins are lower than operating margins).**

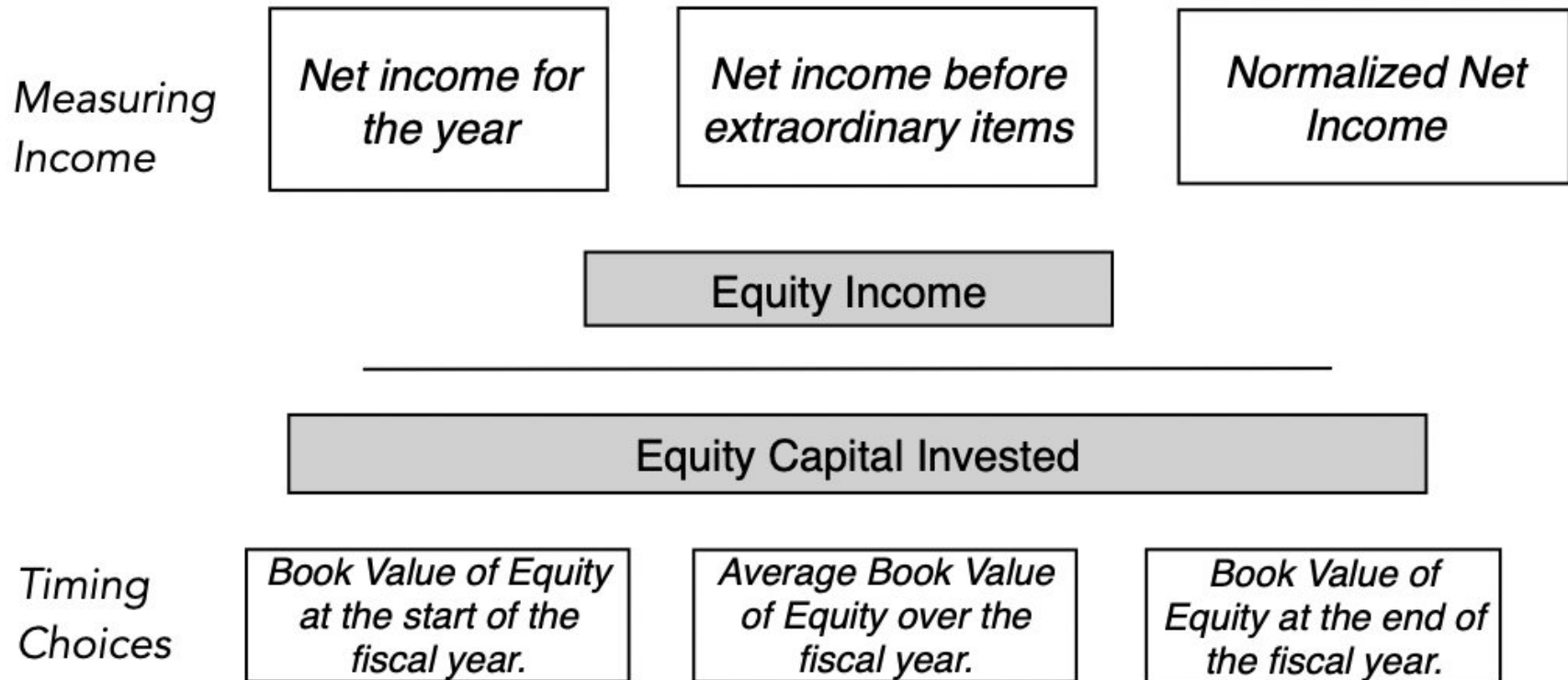
A Life Cycle View of Margins



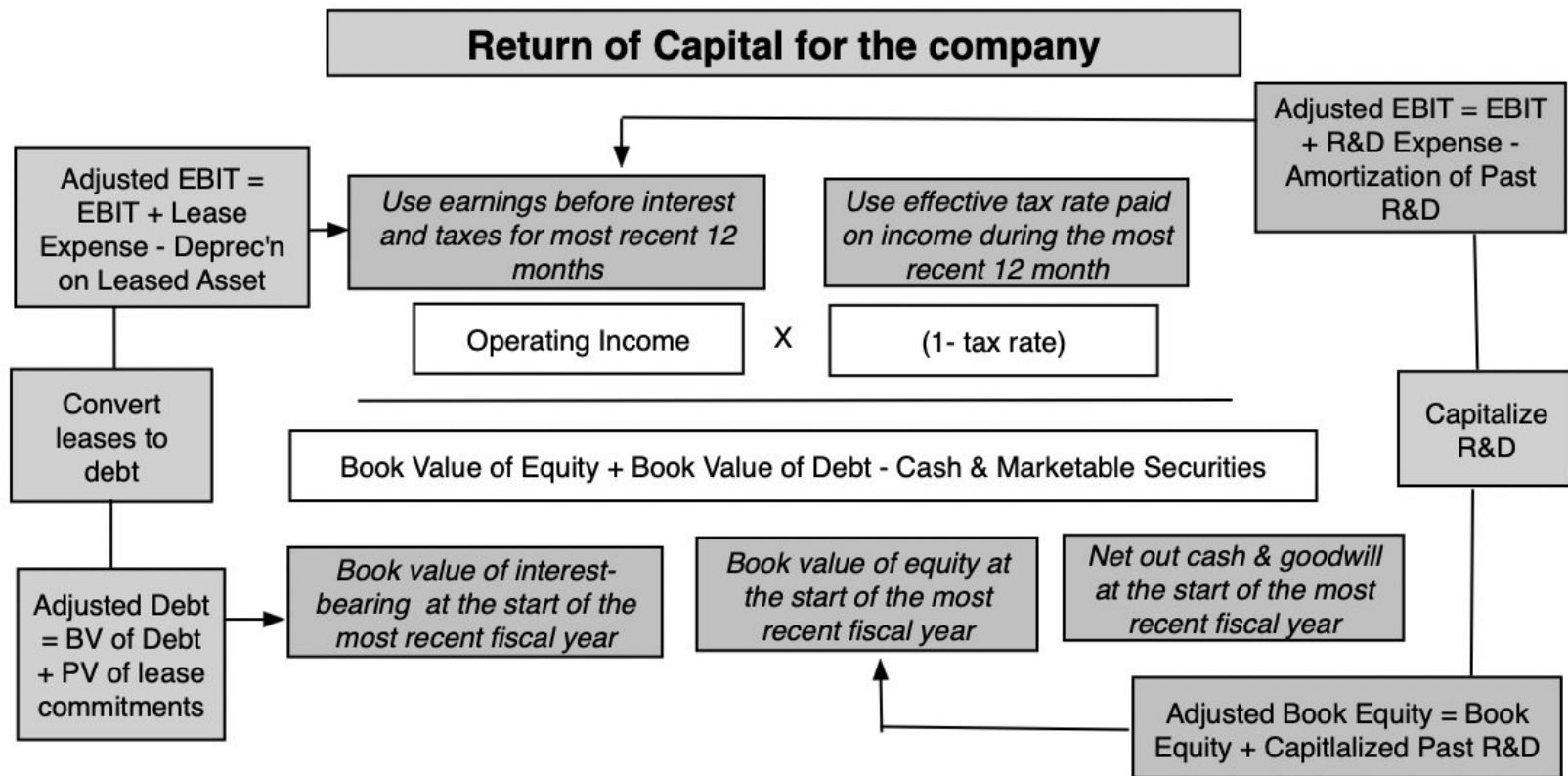
2. Accounting Returns

- With accounting returns, profits are scaled to measures of investment in a project or business.
- Broadly speaking, there can be differences in how accounting returns are measured based upon
 - How profits are measured, i.e., to just equity investors (net income) or to both debt and equity investors and whether profits are before or after taxes. In most cases, it is **accrual income** that is the basis for returns.
 - How investment is measured, i.e., investment made just by equity investors or by debt and equity investors. In most cases, accounting returns **use the book value** as the basis of investment measurement.
 - With any measure of accounting return, you can get different values depending upon timing, i.e., start of the period, end of the period or average for invested capital.
- **Consistency rule:** A consistent measure of accounting return will measure both profits and investment to the same group (equity or capital).

Return on Equity



Return on Invested Capital



3. Efficiency Ratios

- Efficiency ratios measure the revenue payoff that companies get from reinvesting back in their businesses.
- Turnover ratios, with revenues in the numerator are the most widely used measured of efficiency, though the denominator can vary:
 - ▣ *Working Capital Turnover* = Sales/ Non-cash Working capital (or individual items of working capital, like inventory or receivables)
 - ▣ *Asset Turnover* = Sales/ Total Assets
 - ▣ *Capital Turnover* = Sales/ Invested Capital

4. Measuring Financial Leverage

- Debt Ratios measure how much a company has borrowed, relative to overall capital or to earnings/cashflows.
- Debt can be scaled to overall capital or just to equity
 - ▣ Debt to Capital = $\text{Debt} / (\text{Debt} + \text{Equity})$: This is a measure of how much of the capital in a company comes from debt.
 - ▣ Debt to Equity = $\text{Debt} / \text{Equity}$: This is a close variant of debt to capital, with debt stated as a percent of equity.
- Debt can also be measured relative to earnings/cashflows:
 - ▣ Debt to EBITDA = $\text{Debt} / \text{EBITDA}$: This measures how much debt a company has relative to the cash it generates from operations, before taxes and capital expenditures.

Variants on calculation

□ **What to include in debt**

- ▣ Only long term debt
- ▣ All interest bearing debt
- ▣ Debt inclusive of commitments (like leases)

□ **Book or Market**

- ▣ **Book values for debt and equity (from balance sheet)**
- ▣ Market values, measured as market cap for equity in a publicly traded firm, and if doable, market value of debt

□ **Gross or Net**

- ▣ Gross debt is all debt
- ▣ Net debt is all debt minus cash & marketable securities

5. Measuring Liquidity/ Credit Risk

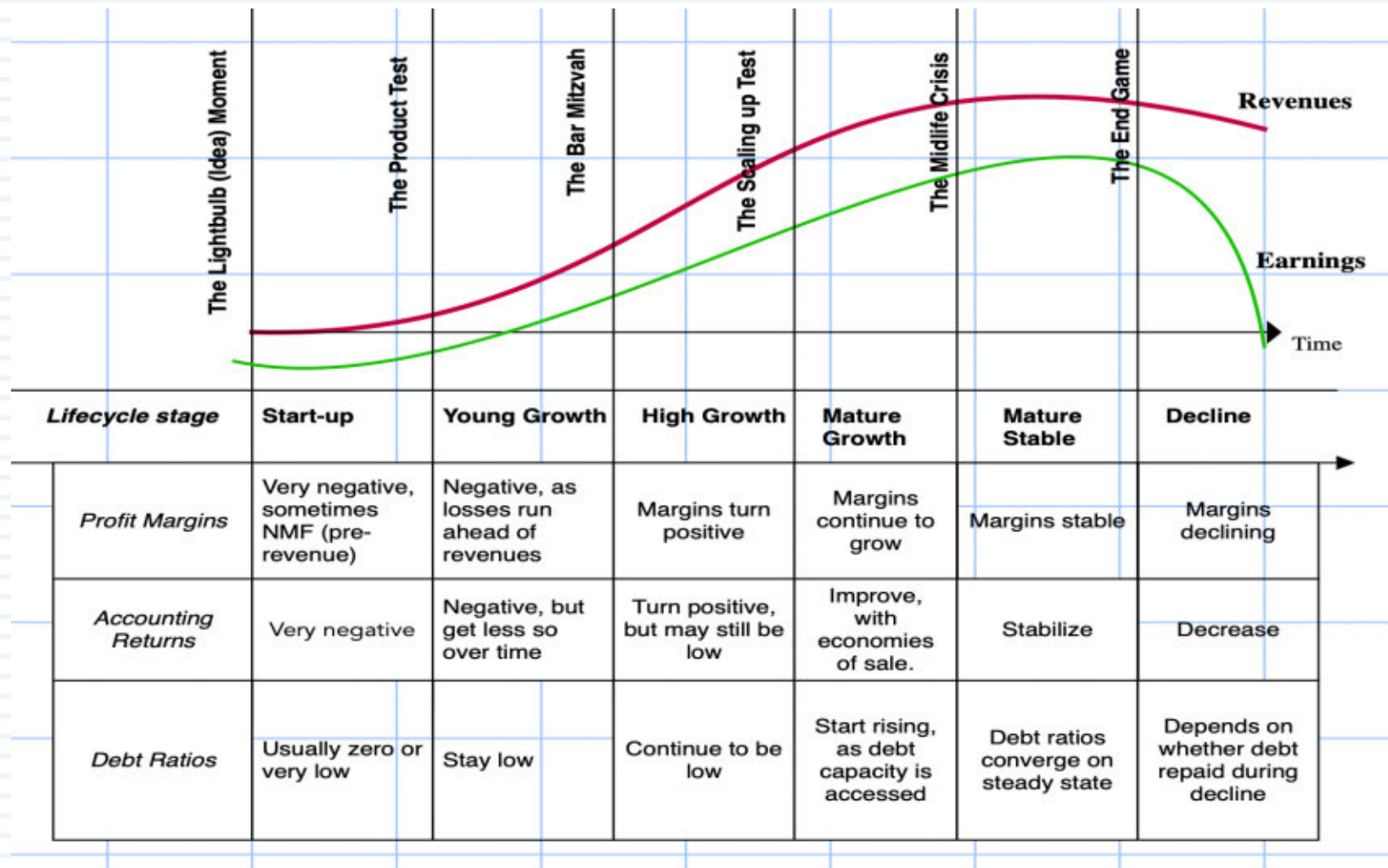
- Coverage Ratios: These ratios measure how much buffer or coverage a company has in meeting commitments.
 - ▣ With interest coverage ratio, the commitment is interest expenses, and it is scaled to operating income.
 - ▣ With a fixed charge coverage ratio, the commitment is expanded to include debt payments, and it is scaled to operating income + fixed charges.
- Liquidity Ratios: These ratios measure how much liquidity companies have, to cover near-term needs or expenses:
 - ▣ Current ratio, measure current assets relative to current liabilities.
 - ▣ Quick ratio, looks at only liquid current assets relative to current liabilities. (Inventory is usually excluded.)



SESSION 6A: RATIO ANALYSIS

Accounting for Finance

Financial Ratios: A Life Cycle Perspective



1. Profitability Ratios

	<i>Peloton</i>	<i>Netflix</i>	<i>Coca Cola</i>	<i>Toyota</i>	<i>Total</i>	<i>Dr. Reddy's</i>
<i>In Income Statement</i>						
Revenues	\$ 219	\$ 20,156	\$ 37,266	¥30,225,681	\$ 200,316	₹ 125,936
Gross Profit	\$ 74	\$ 7,716	\$ 22,647	¥6,836,186	\$ 60,028	₹ 100,371
Operating Income	\$ (71)	\$ 2,605	\$ 10,086	¥2,467,545	\$ 16,257	₹ 28,216
Net Income	\$ (71)	\$ 1,866	\$ 8,985	¥1,985,587	\$ 11,438	₹ 29,377
Gross Margin	33.81%	38.28%	60.77%	22.62%	29.97%	79.70%
Operating Margin	-32.34%	12.92%	27.06%	8.16%	8.12%	22.41%
After-tax Operating Margin	-32.34%	11.70%	22.55%	5.81%	5.29%	23.71%
Net Margin	-32.53%	9.26%	24.11%	6.57%	5.71%	23.33%
Effective Tax Rate	0.00%	9.47%	16.70%	28.88%	34.76%	-5.83%

And analysis...

- Young to old: *Young companies often have negative or very low margins, for two reasons:*
 - ▣ They are still building up their revenues
 - ▣ Some of their operating expenses are associated with future growth, not current operations.
- Business Differences: *High gross margin businesses have the advantage:* Companies like Coca Cola (brand name consumer product) and Dr. Reddy's Labs (pharmaceutical) start with sky-high gross margins, which then feed into high operating and net margins.
- Leverage Effects: *Companies with high debt ratios can have low net margins, while operating margins stay high.*

2. Accounting Returns

	<i>Peloton</i>	<i>Netflix</i>	<i>Coca Cola</i>	<i>Toyota</i>	<i>Total</i>	<i>Dr. Reddy's</i>
<i>Earnings</i>						
Operating Income	\$ (71)	\$ 2,605	\$ 10,086	¥2,467,545	\$ 16,257	₹ 28,216
After-tax Operating Income	\$ (71)	\$ 2,358	\$ 8,402	¥1,755,024	\$ 10,607	₹ 29,862
Net Income	\$ (71)	\$ 1,866	\$ 8,985	¥1,985,587	\$ 11,438	₹ 29,377
<i>Invested Capital (end of previous year)</i>						
Book Equity	\$ 90	\$ 5,239	\$ 19,058	¥20,067,137	\$ 118,114	₹ 126,841
Total Debt	\$ -	\$ 10,360	\$ 44,214	¥15,895,918	\$ 53,435	₹ 8,917
Cash	\$ 151	\$ 3,794	\$ 11,102	¥3,570,704	\$ 27,907	₹ 1,132
Invested Capital	\$ (61)	\$ 11,805	\$ 52,170	¥32,392,351	\$ 143,642	₹ 134,626
<i>Accounting Returns</i>						
Return on Equity	-79.00%	35.62%	47.15%	9.89%	9.68%	23.16%
ROIC	NA	22.07%	19.33%	7.62%	11.32%	20.96%
After-tax ROIC	NA	19.98%	16.10%	5.42%	7.38%	22.18%

Reflections of...

- Competitive Advantages: If accounting returns are fair measure of true returns, they are the repository for competitive advantages.
 - ▣ Industries where barriers to entry are high or other competitive advantages prevail should have higher returns on capital than companies without these advantages.
 - ▣ Companies with strong competitive advantages within an industry should earn higher returns than their peer group.
- Accounting choices and inconsistencies: Accounting can affect and sometimes skew returns:
 - ▣ By misclassifying capital, operating and financial expenses
 - ▣ By taking write offs to reflect mistakes made in the past.

3. Efficiency Ratios

	<i>Peloton</i>	<i>Netflix</i>	<i>Coca Cola</i>	<i>Toyota</i>	<i>Total</i>	<i>Dr. Reddy's</i>
<i>Operating Metrics</i>						
Revenues	\$ 219	\$ 20,156	\$ 37,266	¥30,225,681	\$ 200,316	₹ 125,936
Total Assets (prior year)	\$ 271	\$ 25,974	\$ 83,216	¥51,936,949	\$ 256,762	₹ 162,475
Invested Capital (prior year)	\$ (61)	\$ 11,805	\$ 52,170	¥32,392,351	\$ 143,642	₹ 134,626
<i>Turnover Ratios</i>						
Asset Turnover Ratio	0.81	0.78	0.45	0.58	0.78	0.78
Sales to Invested Capital	NA	1.71	0.71	0.93	1.39	0.94

The dark side of growth

- The growth trade off: Growth has a good side, insofar as it lets a company scale up its operations, but it has a dark side, which is that companies have to reinvest to deliver that growth.
- Scaling up measure: Turnover ratios look at the link between what a company has to reinvest, and how much its revenues grow over time. Companies that are more efficient on this measure will be able to grow revenues, with less reinvestment.
 - Young to old: The link between company age and turnover ratios will vary across different business types, with older companies becoming more efficient in some, and less in others.
 - Accounting effects: The problems associated with accounting choices and inconsistencies will affect turnover ratios as well.

4. Debt Ratios

	<i>Peloton</i>	<i>Netflix</i>	<i>Coca Cola</i>	<i>Toyota</i>	<i>Total</i>	<i>Dr. Reddy's</i>
<i>Operating Metrics</i>						
Long Term Debt	\$ -	\$ 14,759	\$ 27,516	¥10,692,898	\$ 47,773	₹ 193
Total Debt	\$ -	\$ 14,759	\$ 42,763	¥20,552,969	\$ 62,592	₹ 10,629
Net Debt	\$ (162)	\$ 9,741	\$ 34,816	¥16,362,451	\$ 35,240	₹ 10,237
Book Equity	\$ 402	\$ 7,582	\$ 52,170	¥19,348,152	\$119,305	₹ 151,919
Market Equity	\$ 8,100	\$ 216,350	\$205,790	¥19,753,000	\$104,610	₹ 746,383
EBITDA	\$ (181)	\$ 11,924	\$ 11,451	¥4,259,920	\$ 32,658	₹ 36,108
<i>Turnover Ratios</i>						
Debt to Equity (Book)	0.00%	194.66%	81.97%	106.23%	52.46%	7.00%
Debt to Equity (Market)	0.00%	6.82%	20.78%	104.05%	59.83%	1.42%
Debt to Capital (Book)	0.00%	66.06%	45.05%	51.51%	34.41%	6.54%
Debt to Capital (Market)	0.00%	6.39%	17.20%	50.99%	37.43%	1.40%
Debt to EBITDA	NA	1.24	3.73	4.82	1.92	0.29
Net Debt to EBITDA	NA	0.82	3.04	3.84	1.08	0.28

Debt, the double edged sword

- Source of capital: Debt is a source of capital for a business, just as equity is. There is nothing inherently good or bad about it, but in most parts of the world, it is a trade off between tax benefits that accrue to borrowing and distress risk.
- Measurement choices: When comparing across companies, you have to measure debt consistently across companies. However, it is usually better to focus on
 - ▣ Total debt, rather than a subset of debt
 - ▣ Market value, rather than book value
- Gross vs Net Debt: While the rationale for netting cash out from debt is impeccable, cash is a transient asset, here today and can be gone tomorrow.

5. Coverage & Liquidity Ratios

	<i>Peloton</i>	<i>Netflix</i>	<i>Coca Cola</i>	<i>Toyota</i>	<i>Total</i>	<i>Dr. Reddy's</i>
<i>Operating Metrics</i>						
Interest Expenses	\$ -	\$ 626	\$ 946	¥32,217	\$ 2,333	₹ 527
Debt Repayments	\$ -	\$ -	\$ 24,850	¥4,442,332	\$ 5,829	₹ 1,805
Operating income	\$ (71)	\$ 2,605	\$ 10,086	¥2,467,545	\$ 16,257	₹ 28,216
<i>Turnover Ratios</i>						
Inerest Coverage Ratio	NA	4.16	10.66	76.59	6.97	53.54
Fixed Charge Coverage	NA	5.16	1.39	1.55	2.99	13.10
Current Assets	\$ 582	\$ 6,178	\$ 20,411	¥18,642,531	\$ 85,265	₹ 101,067
Current Liabilities	\$ 291	\$ 6,856	\$ 26,973	¥17,902,377	\$ 70,244	₹ 41,805
Current Ratio	2.00	0.90	0.76	1.04	1.21	2.42
Current Assets without inventory	\$ 445	\$ 6,178	\$ 17,032	¥16,207,613	\$ 68,133	₹ 79,163
Current Liabilities	\$ 291.00	\$ 6,856	\$ 26,973	¥17,902,377	₹ 70,244	₹ 41,805
Quick Ratio	1.53	0.90	0.63	0.91	0.97	1.89

And analysis...

- Safety in numbers: All else held equal, companies that score higher on interest and fixed charge coverage ratios should have more buffer than companies that score lower.
- Normalization? That said, the ratios can be skewed by year-to-year changes, especially in operating income and debt repayments.
 - For companies in volatile businesses, this can translate into big swings in coverage ratios from good to bad years. The solution is to use an average across time.
 - For young companies, the coverage ratios can look bad, at least as they start the growth process, but these companies can grow operating income quickly to gain buffers.

Final Thoughts

- Less is more: If you decide to use financial ratios, less is more. Choose the ratios that you want, rather than create noise by computing multiple ratios.
- A means to an end: Ratios, by themselves, are just numbers and mean nothing, unless you use them to make judgments about what your company does well or badly, and how this affects your perspective for the company.
- Past versus future: While your ratios are in the past, investing and corporate finance are about the future.